



DIPARTIMENTO DI FISICA
SAPIENZA
UNIVERSITÀ DI ROMA

Spin glass models are good benchmarks for discrete combinatorial optimization

Federico Ricci-Tersenghi
(Sapienza University of Rome)

BENQODHI workshop
10–11 September 2026 · JRC Ispra

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- **Widespread problems** in academic research and real-world applications

It is likely to find COP in real-life applications

- Quadratic Assignment Problem
- Number Partition Problem
- Scheduling Problem
- Knapsack Problem
- Vehicle Routing Problem
- Portfolio Optimization
- Exams/Lectures Timetabling
- ...

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- Physics-inspired improved algorithms
- How to prove a new algorithm is “better” than existing ones?
→ Robust benchmarks (possibly with certified solutions)

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- May include among the hardest COP
randomness → lack of correlation/structure the algorithm can exploit

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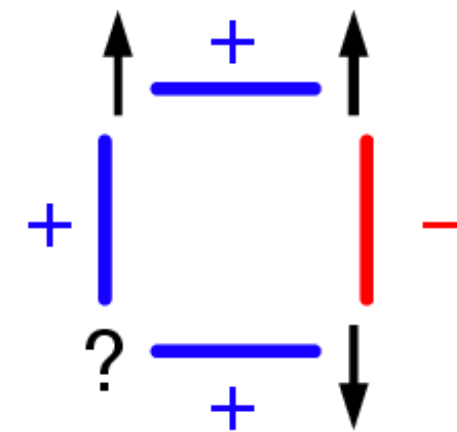
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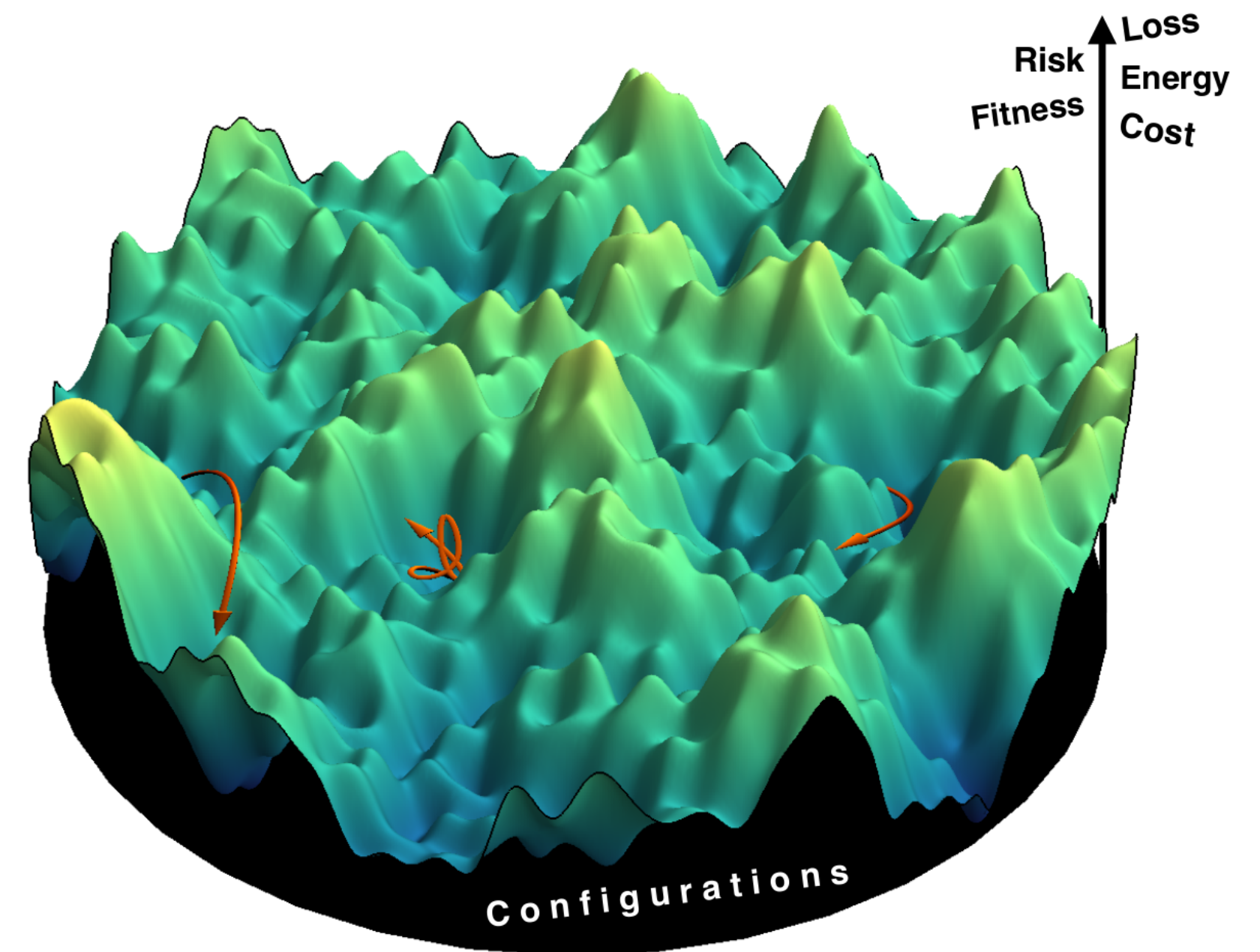
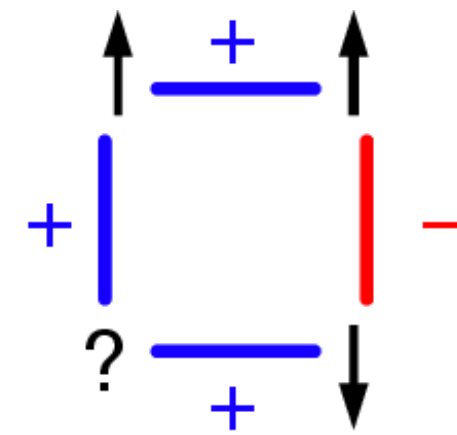
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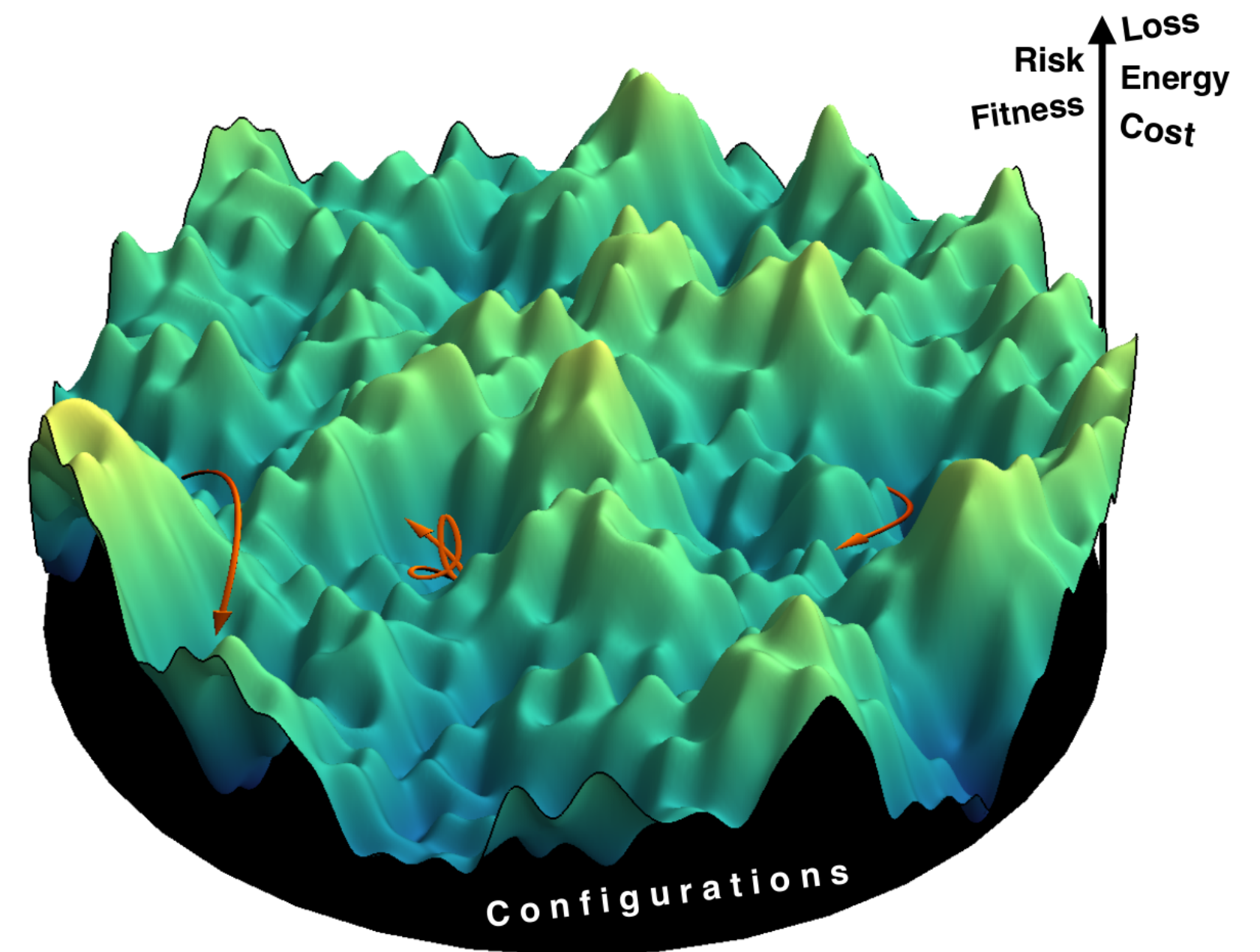
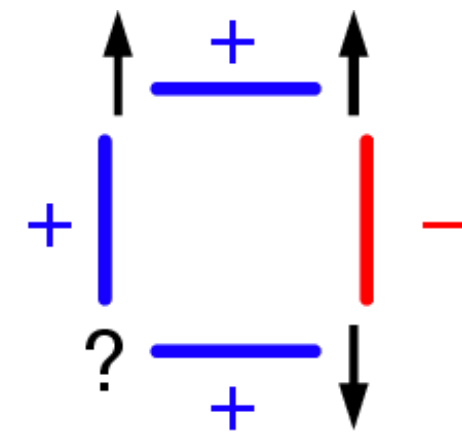
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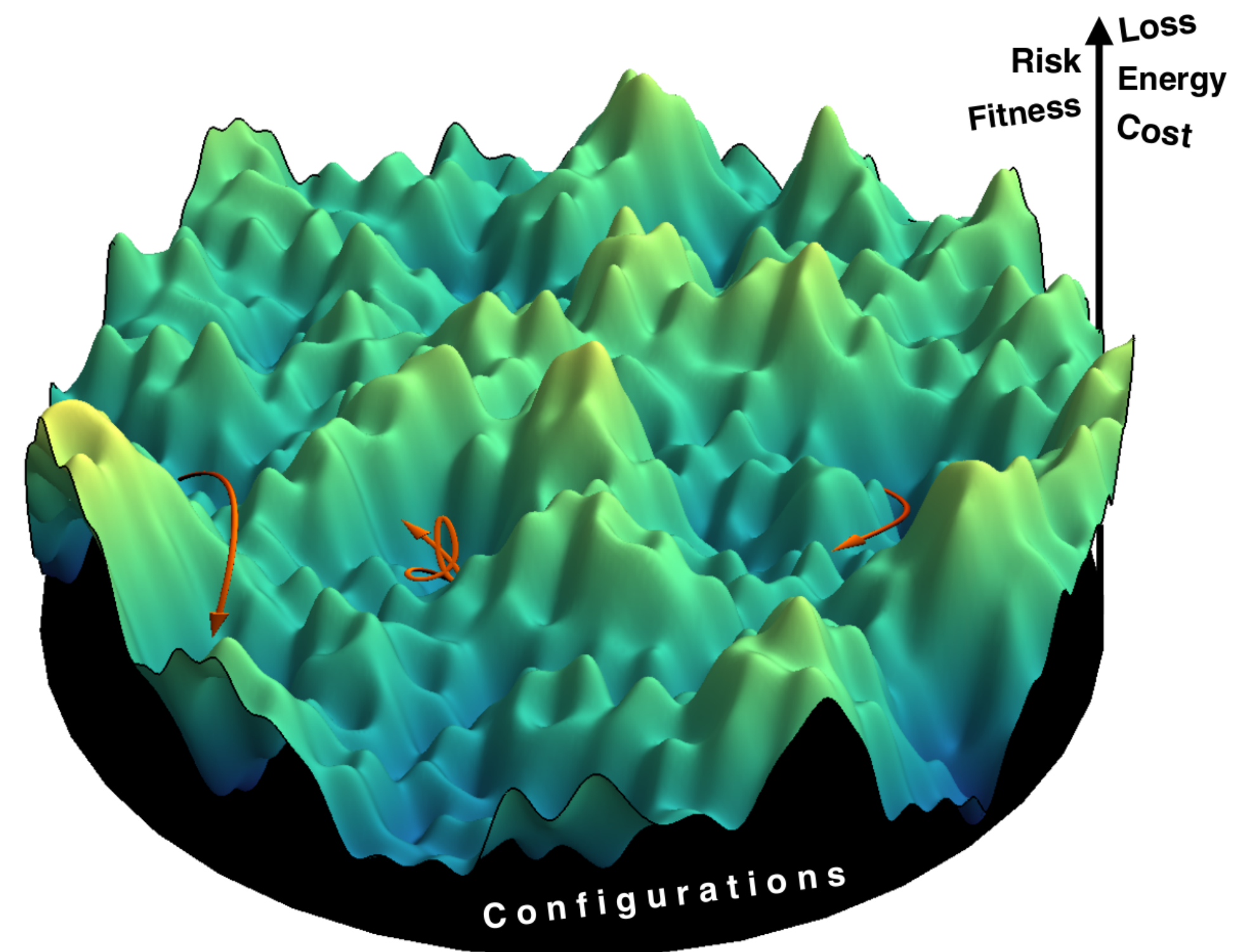
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- **Energetic and entropic barriers** trap algorithms looking for the ground state



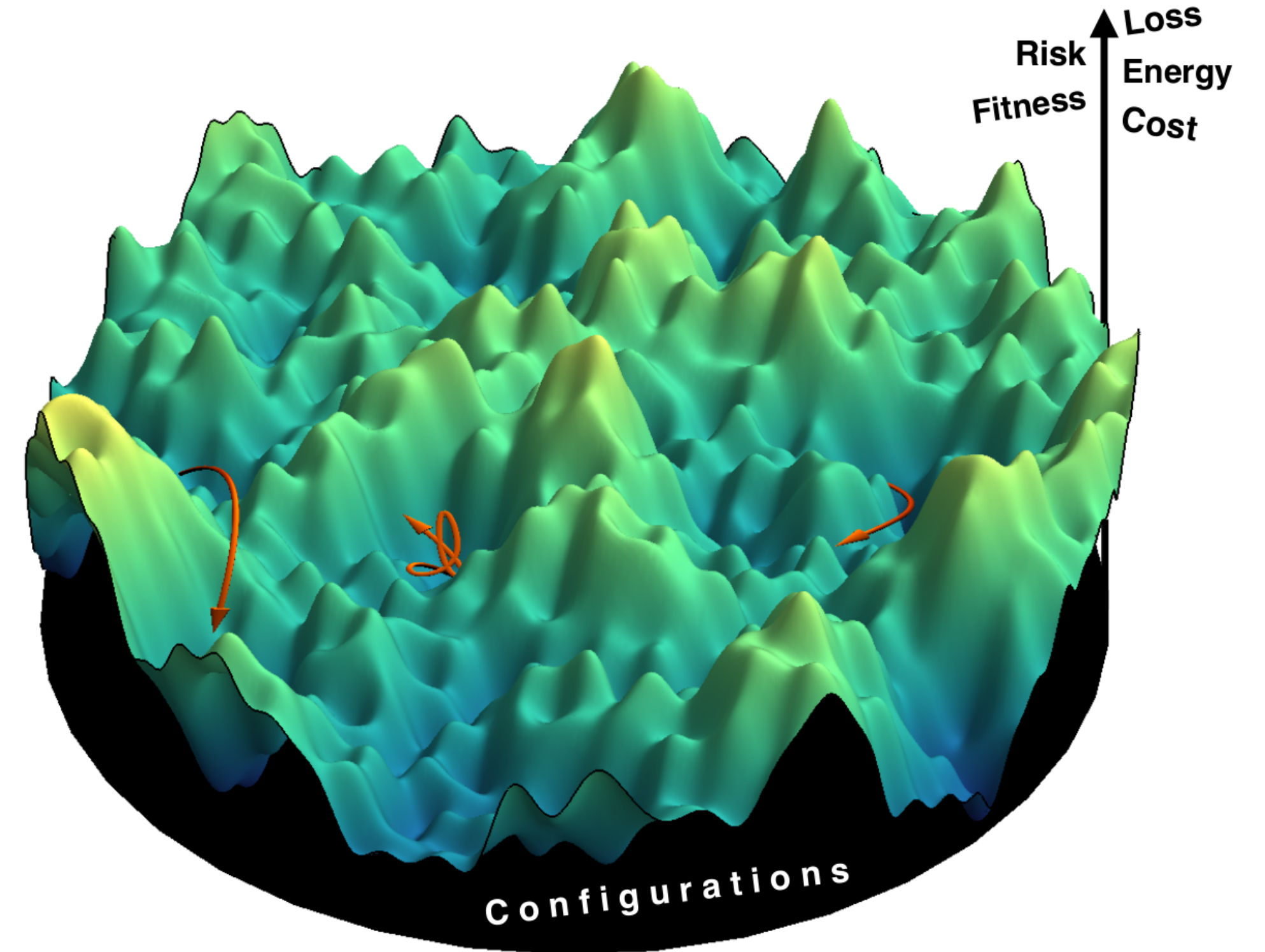
Statistical physics of disordered systems



**Curse of
dimensionality**

Statistical physics of disordered systems

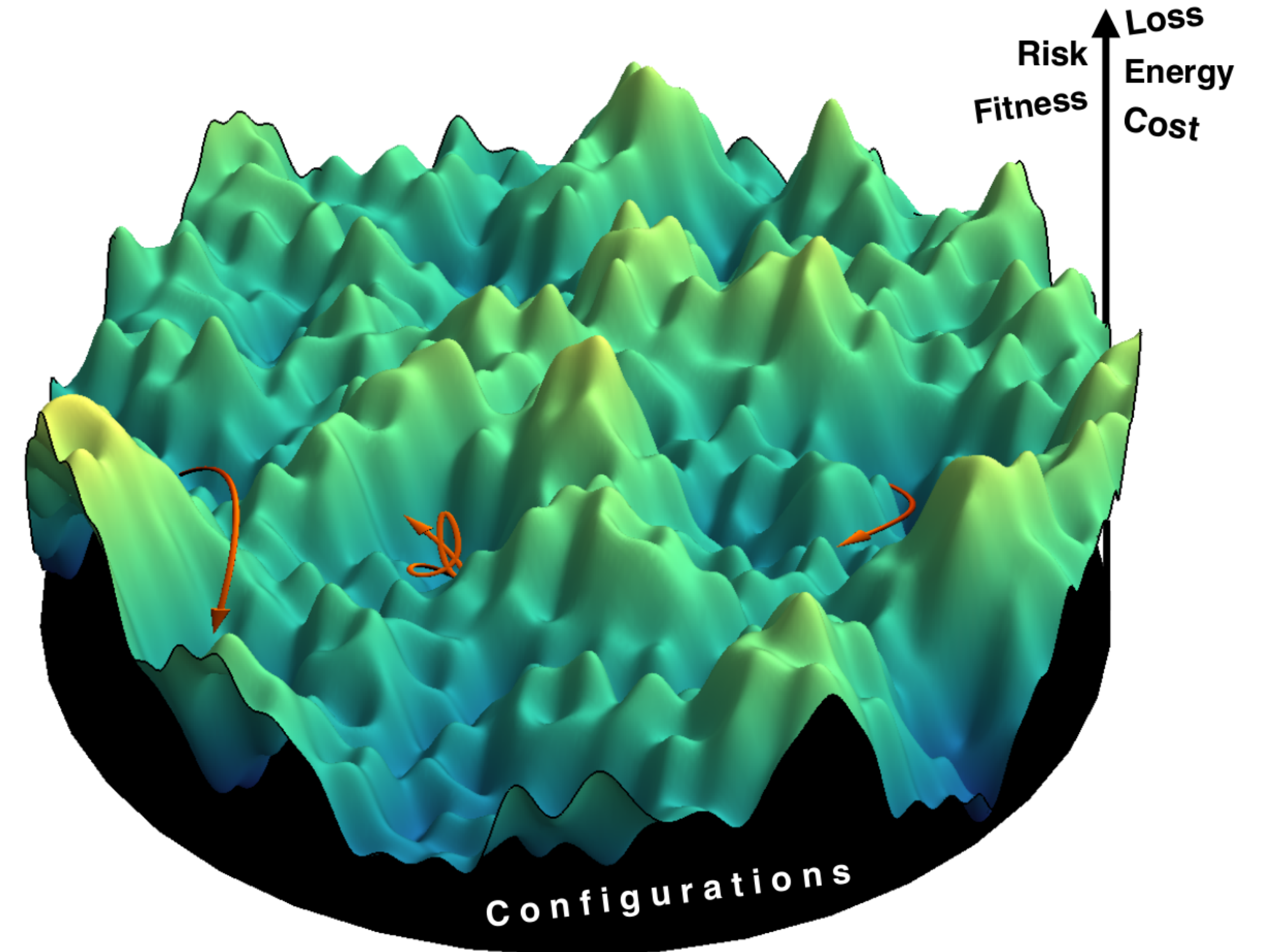
- Very powerful tools in the large N limit
 - Replica Symmetry Breaking theory (2021 Nobel in Physics to Giorgio Parisi)
 - Cavity method
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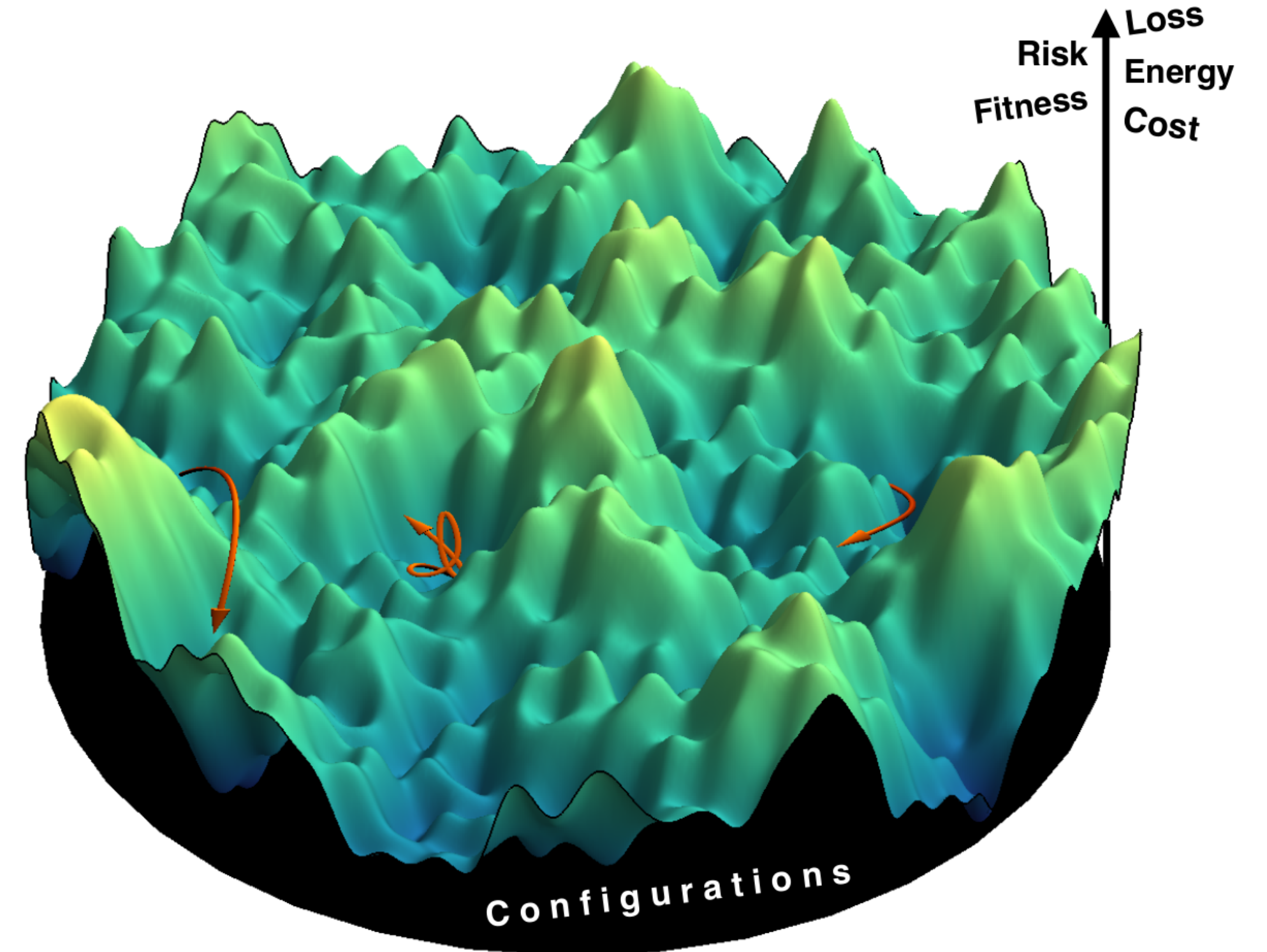
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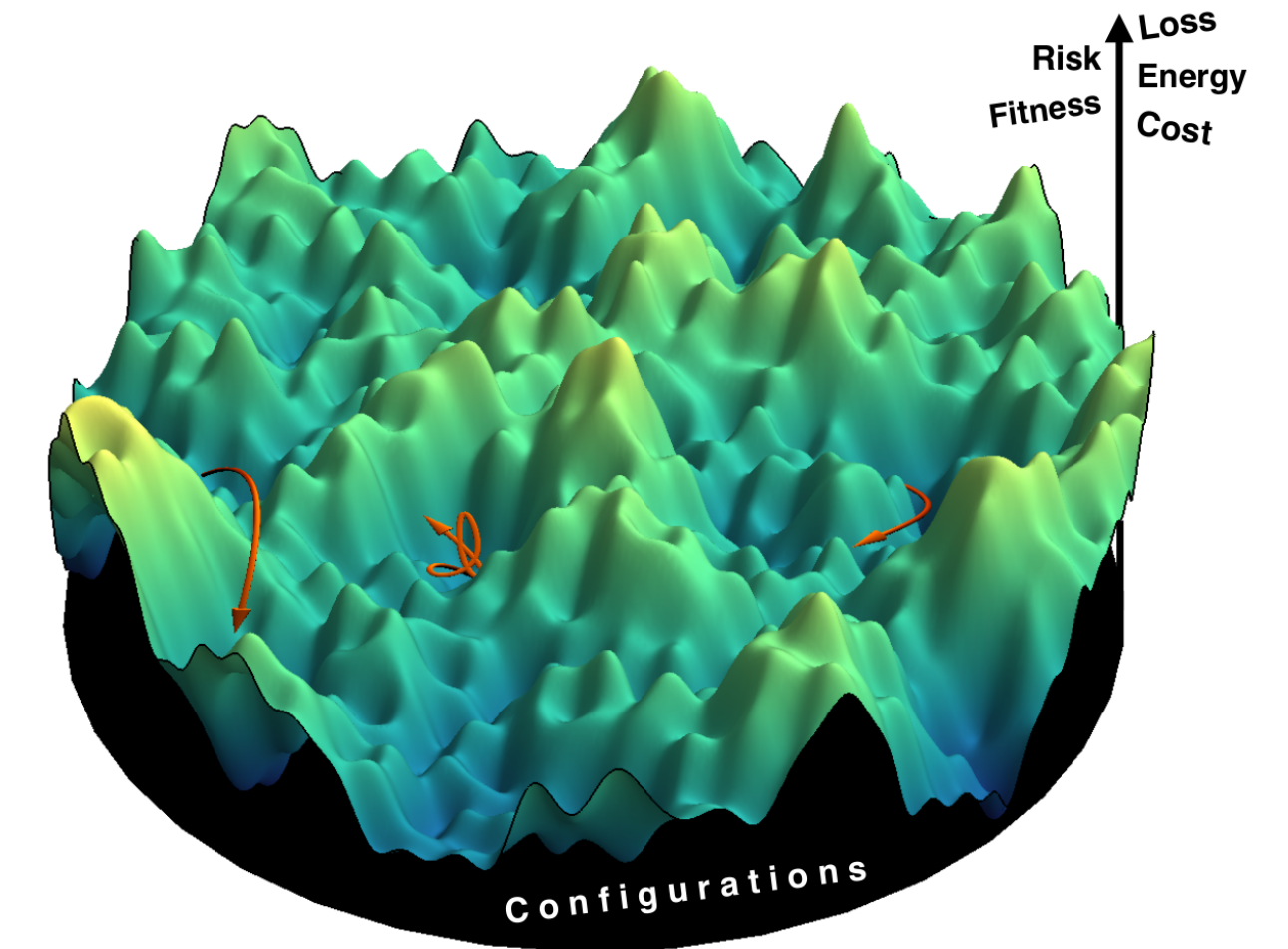


**Blessing of
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Spin glass models as a good benchmark for COP solvers

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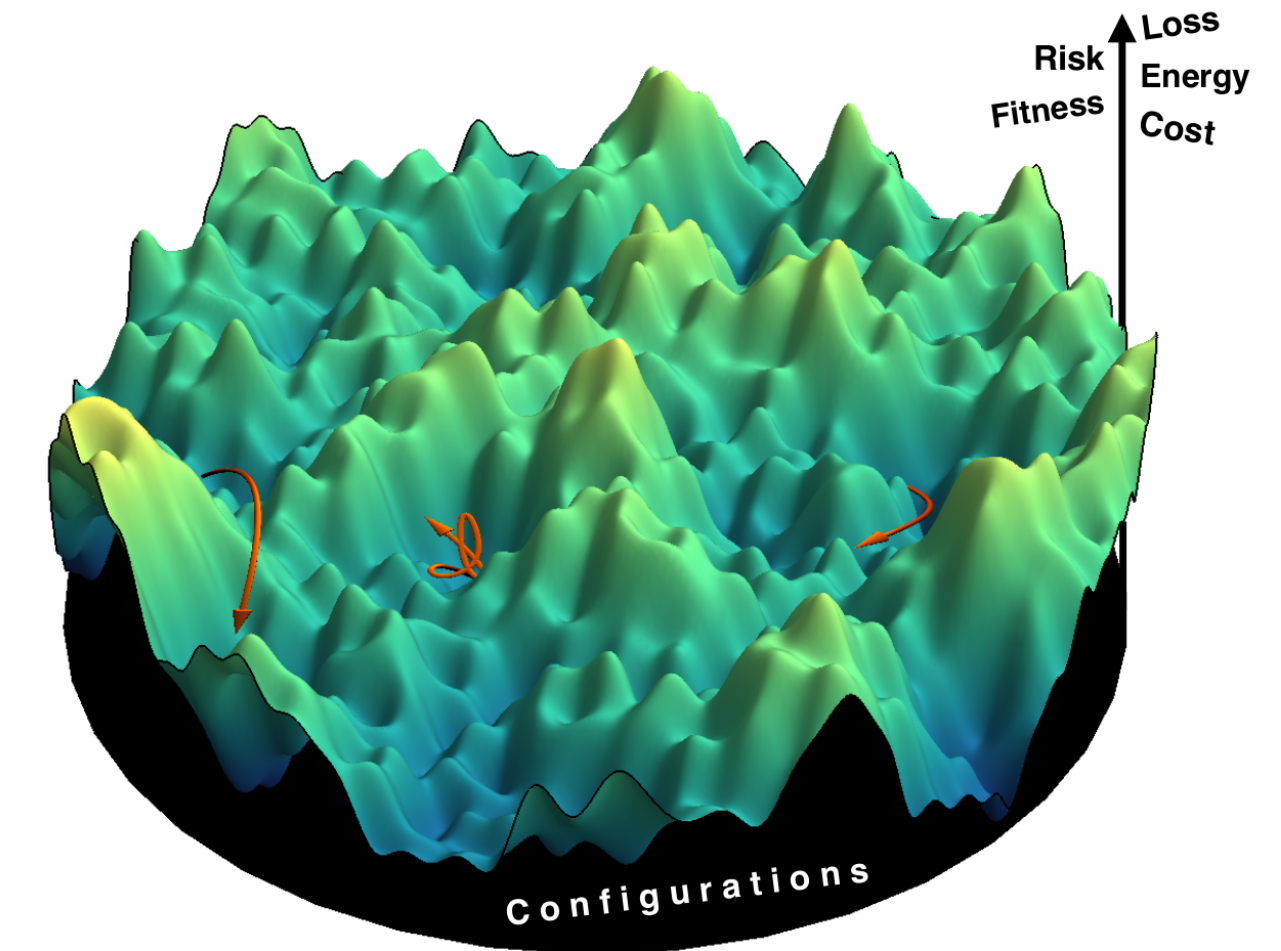


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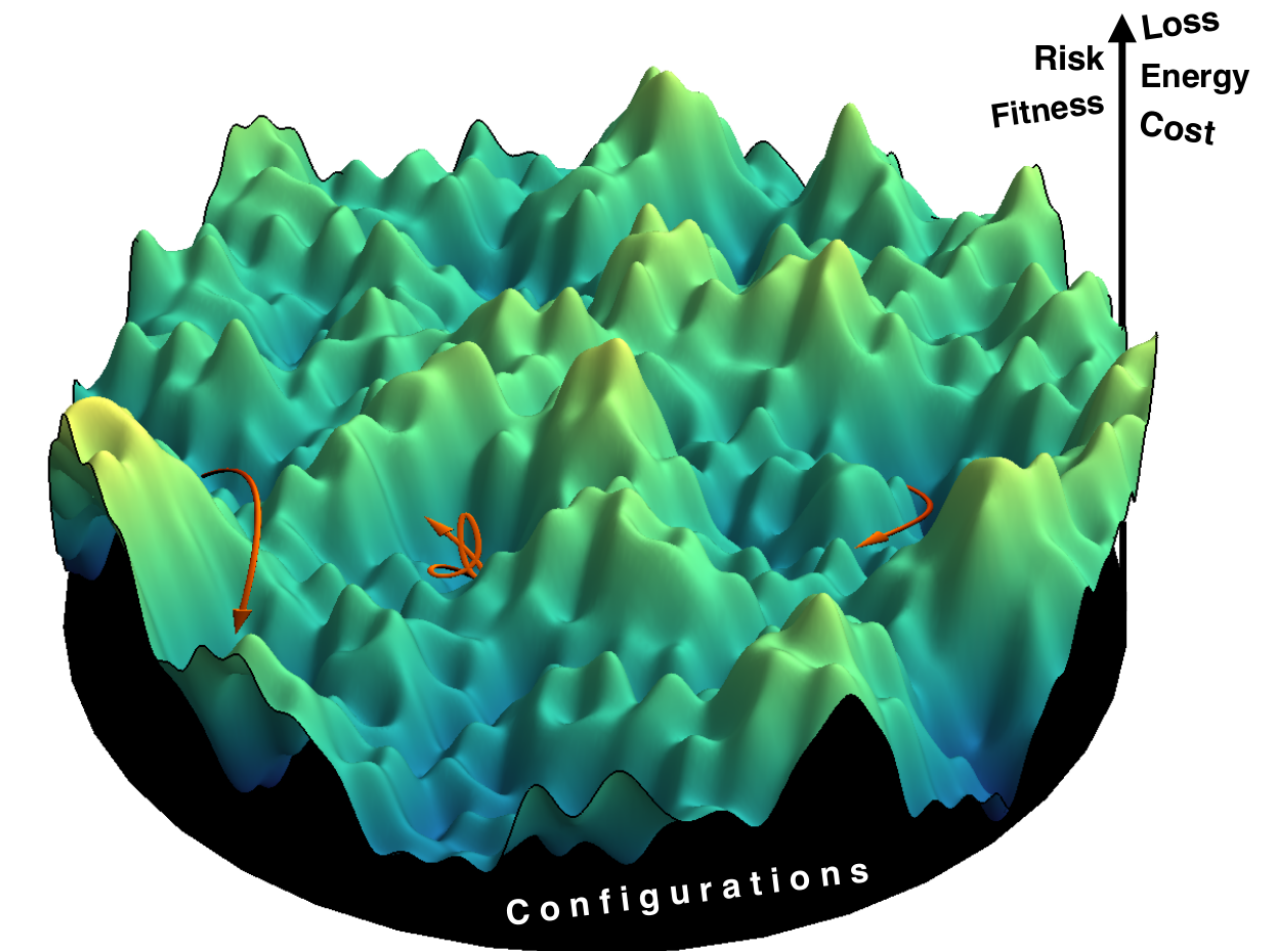


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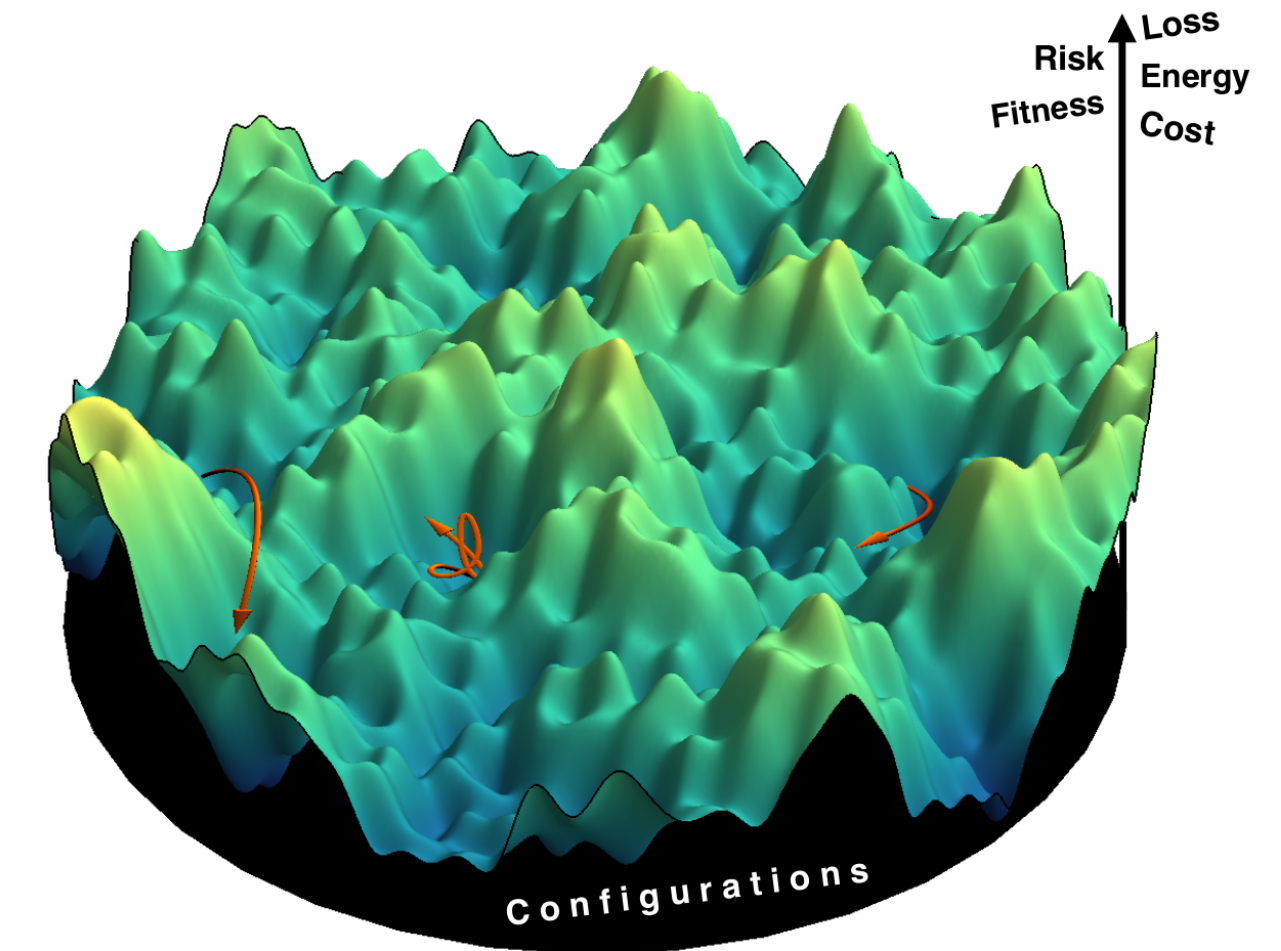


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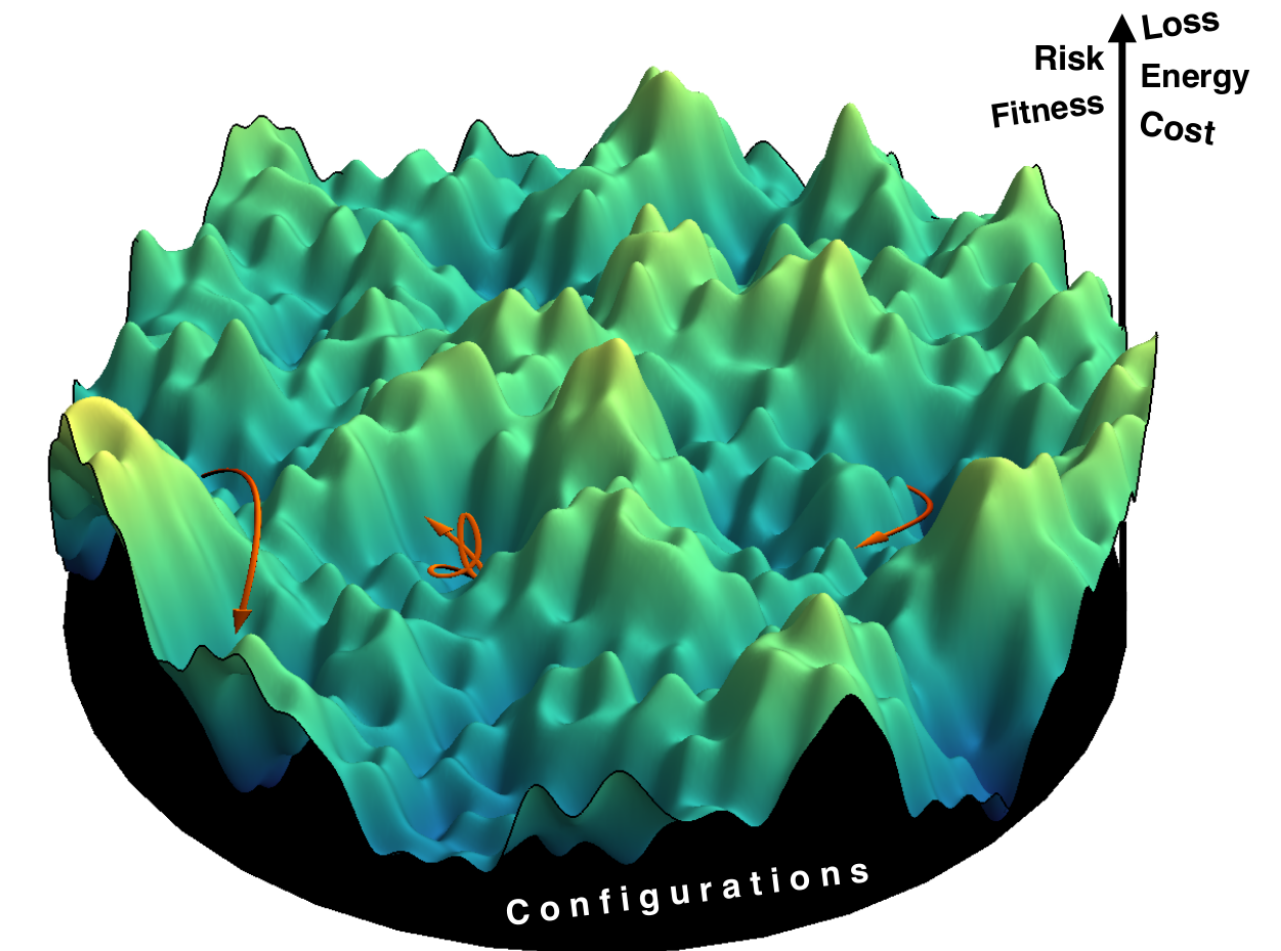


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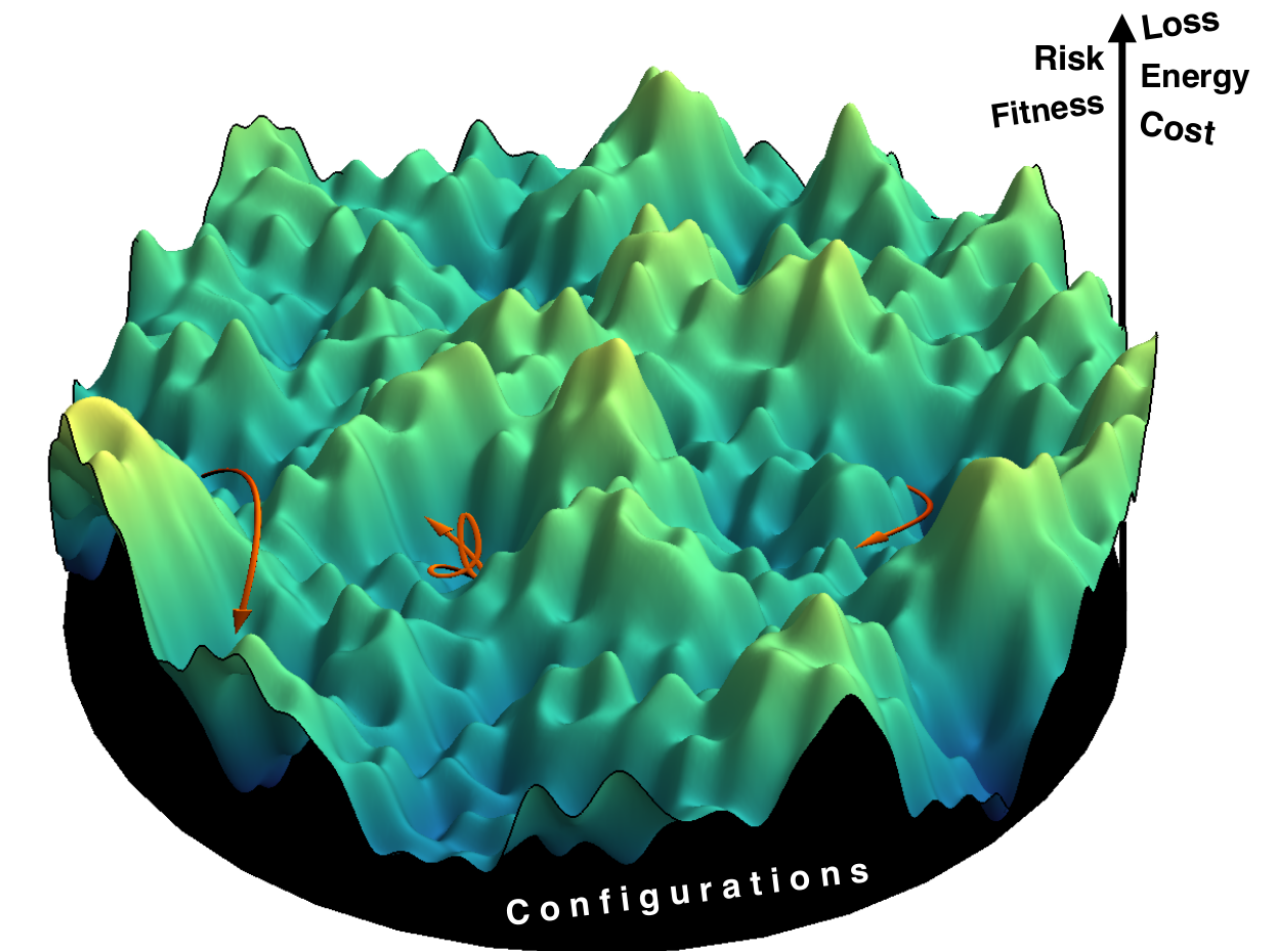
Afternoon talk
by Francesco
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- Be aware of temperature chaos



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- FPGA-based digital annealer
 - coherent Ising machine
 - MemComputing machine
 - quantum annealer

Two types of spin glass models

- **Continuous** phase transition

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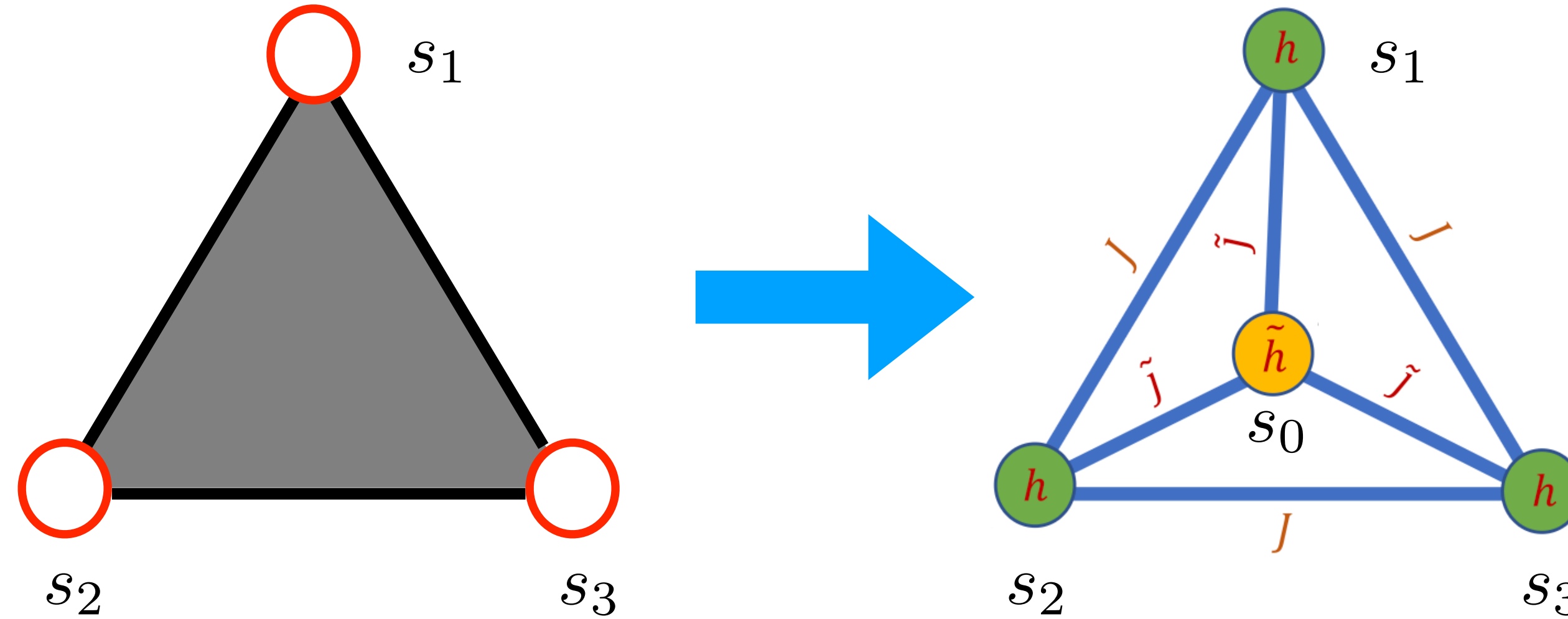
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interaction
topology

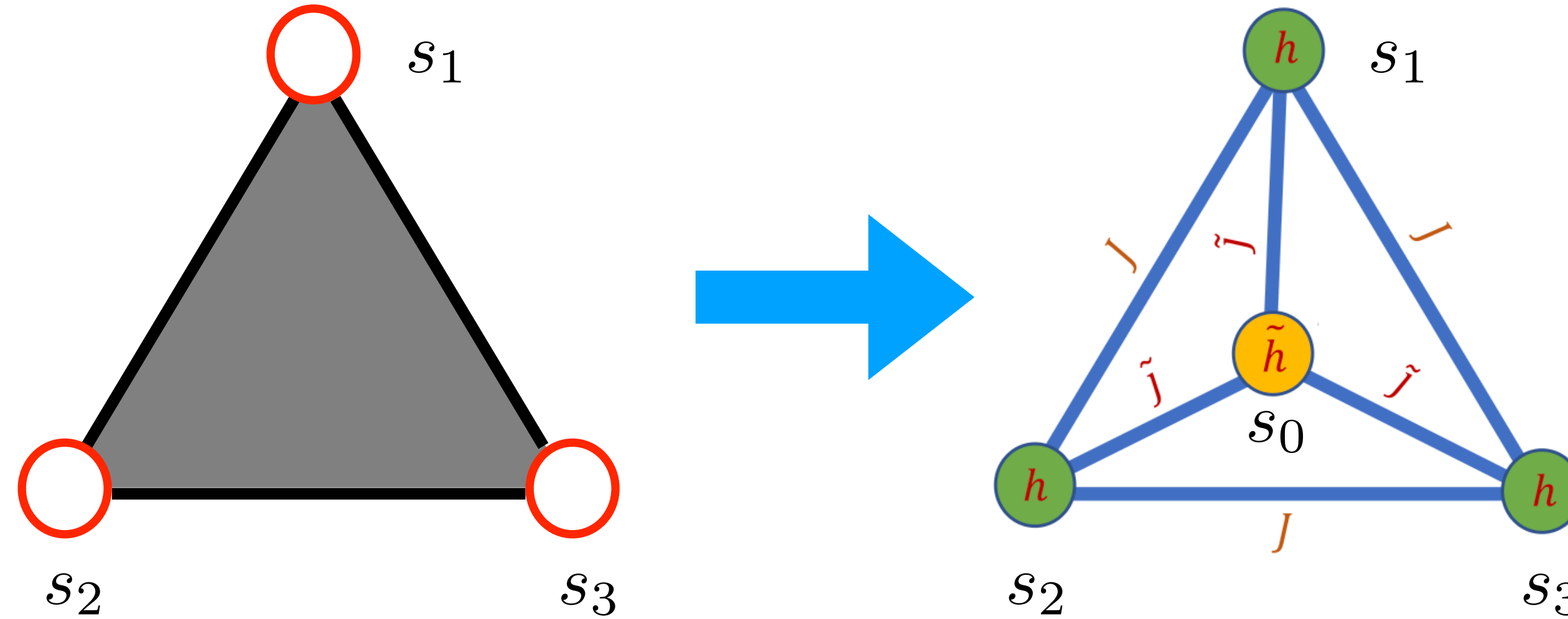
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From the 3-spin model to a 2-spin equivalent model



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The 3-spin model has i.i.d. random couplings

The corresponding 2-spin model in a field has
correlated disorder (i.e. couplings and fields)

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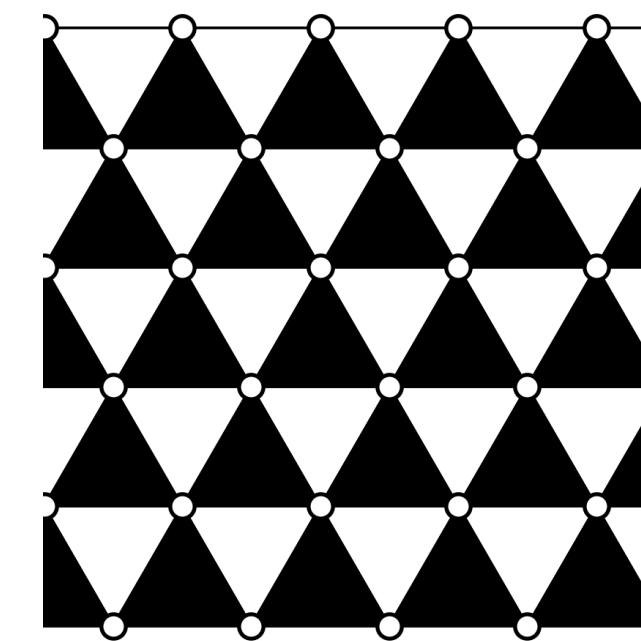
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 - A good model for glassy physics
 - Random First Order Transition (RFOT)
 - Very slow dynamical evolution at low temperatures
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- In a **2D geometry**
 - Trivial thermodynamics
 - No phase transition
 - Easy to reach the (degenerate) ground state

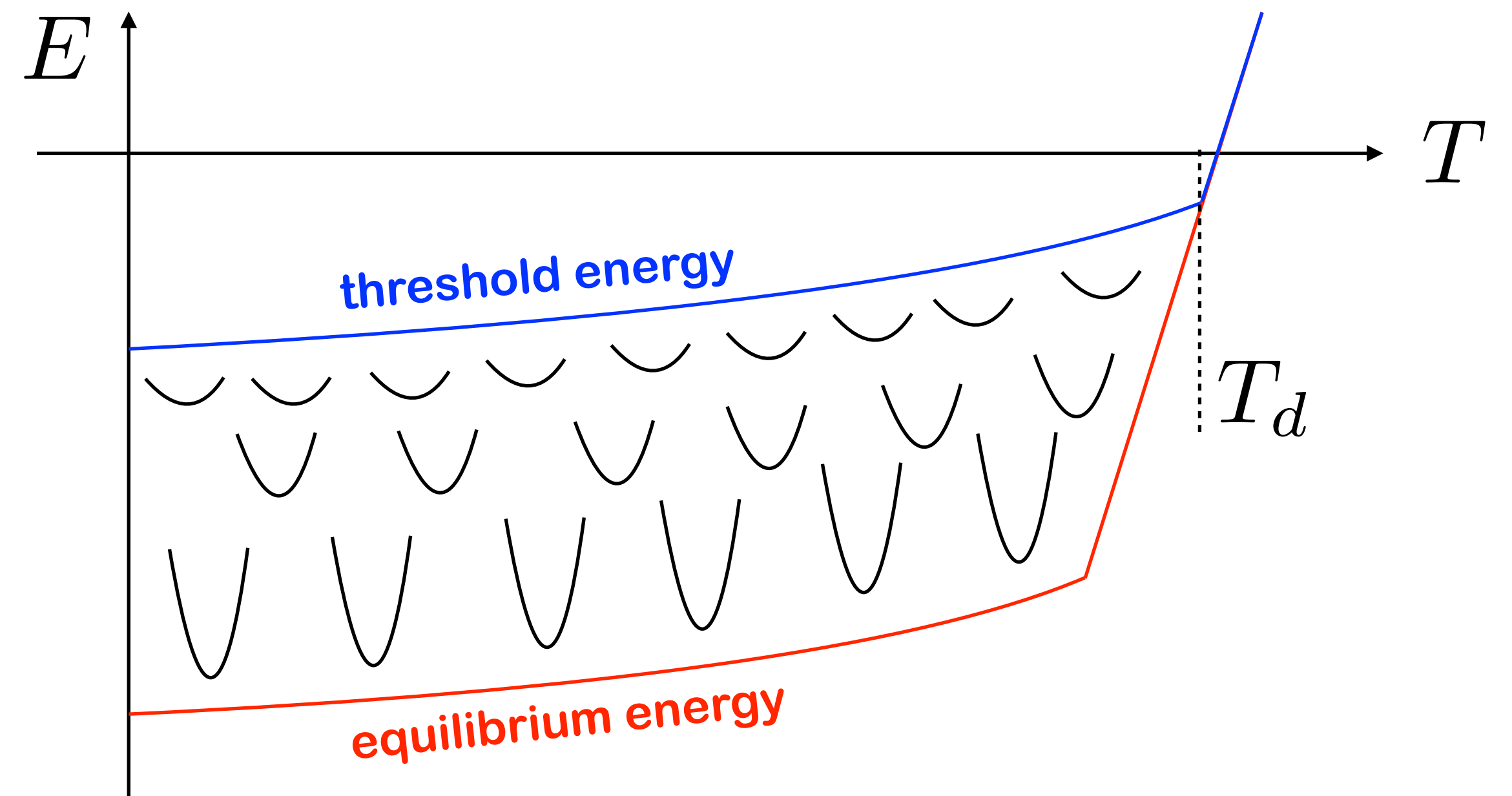


Energy landscape of p-spin glass models

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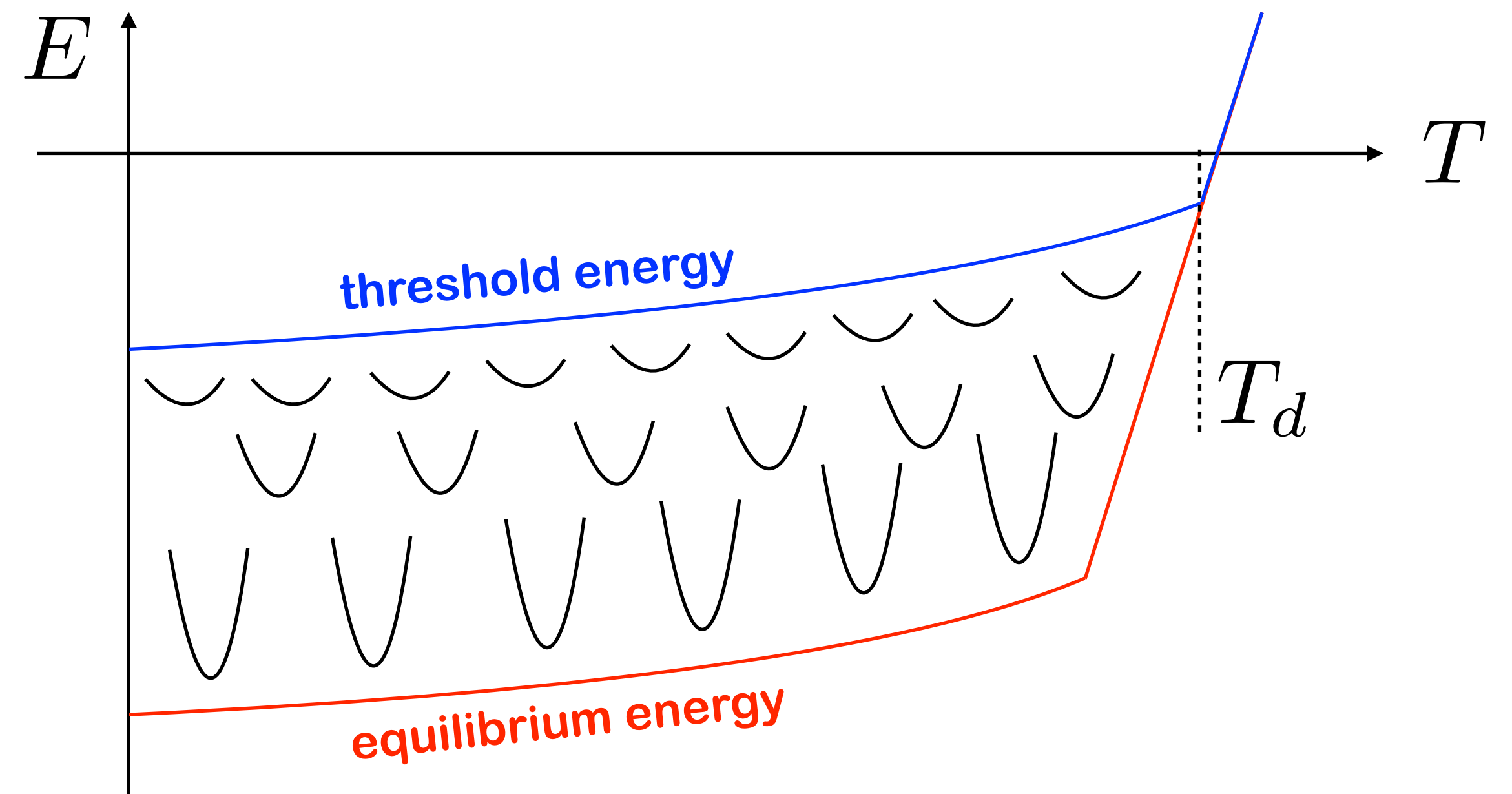


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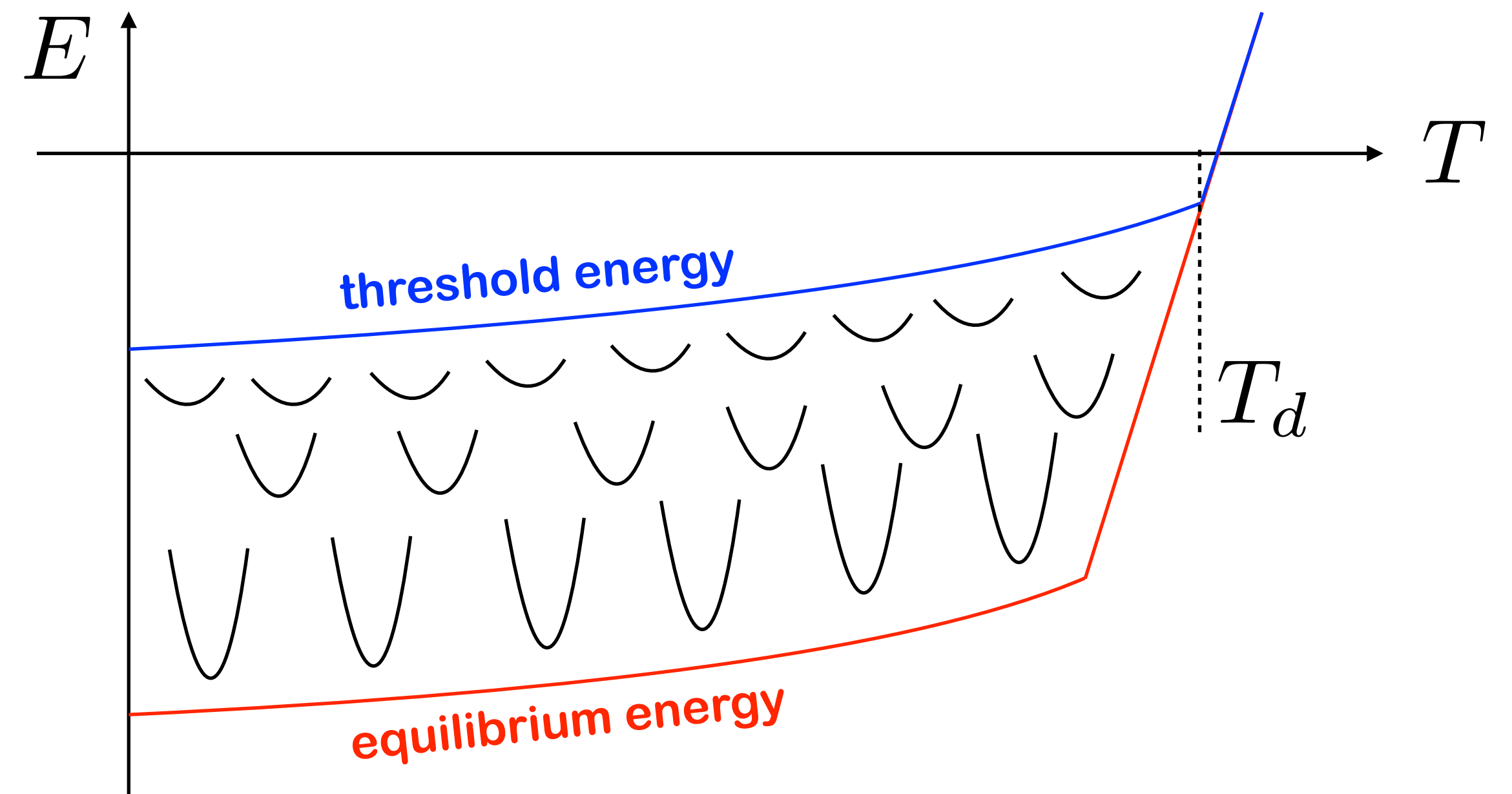


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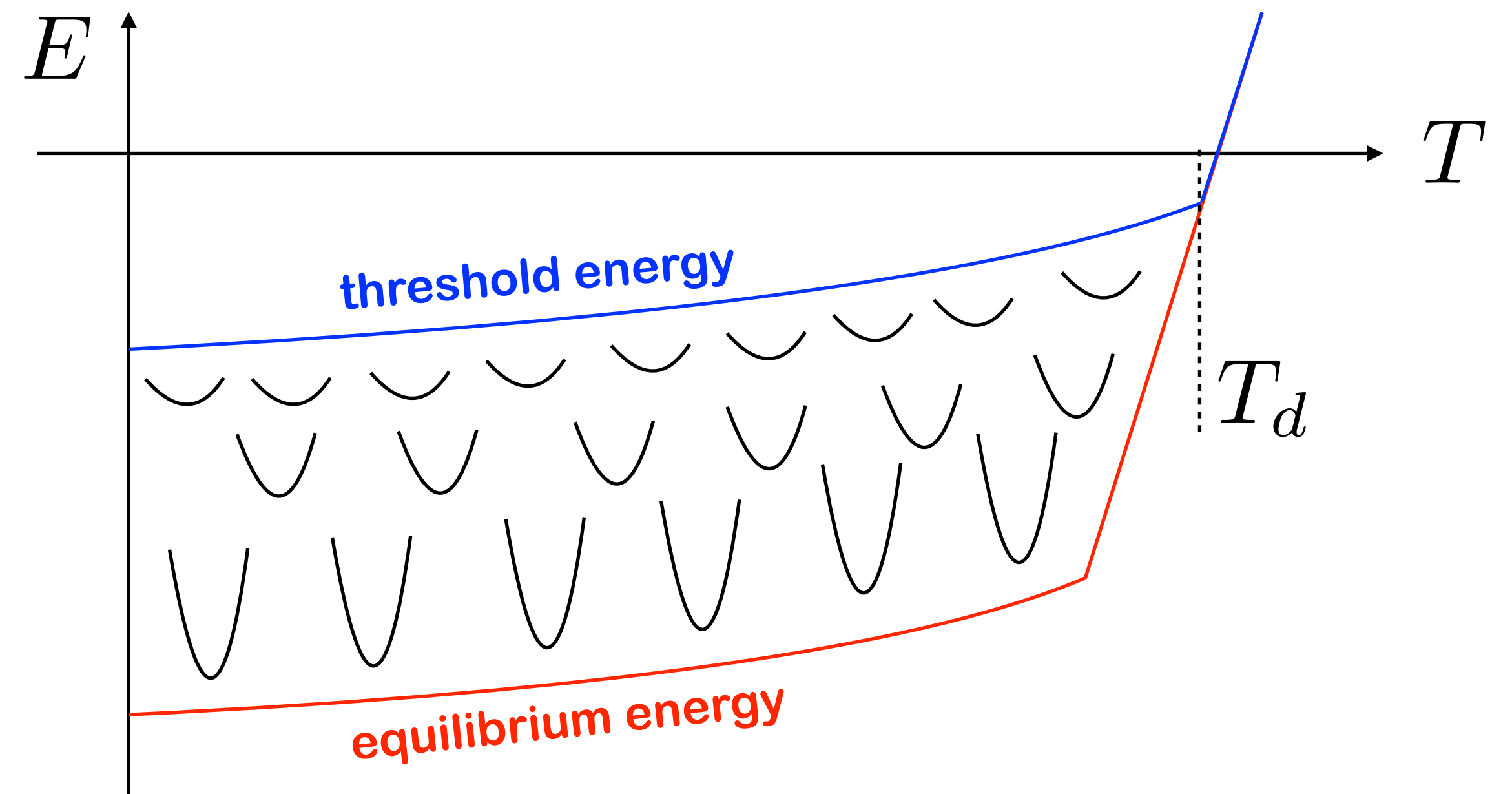
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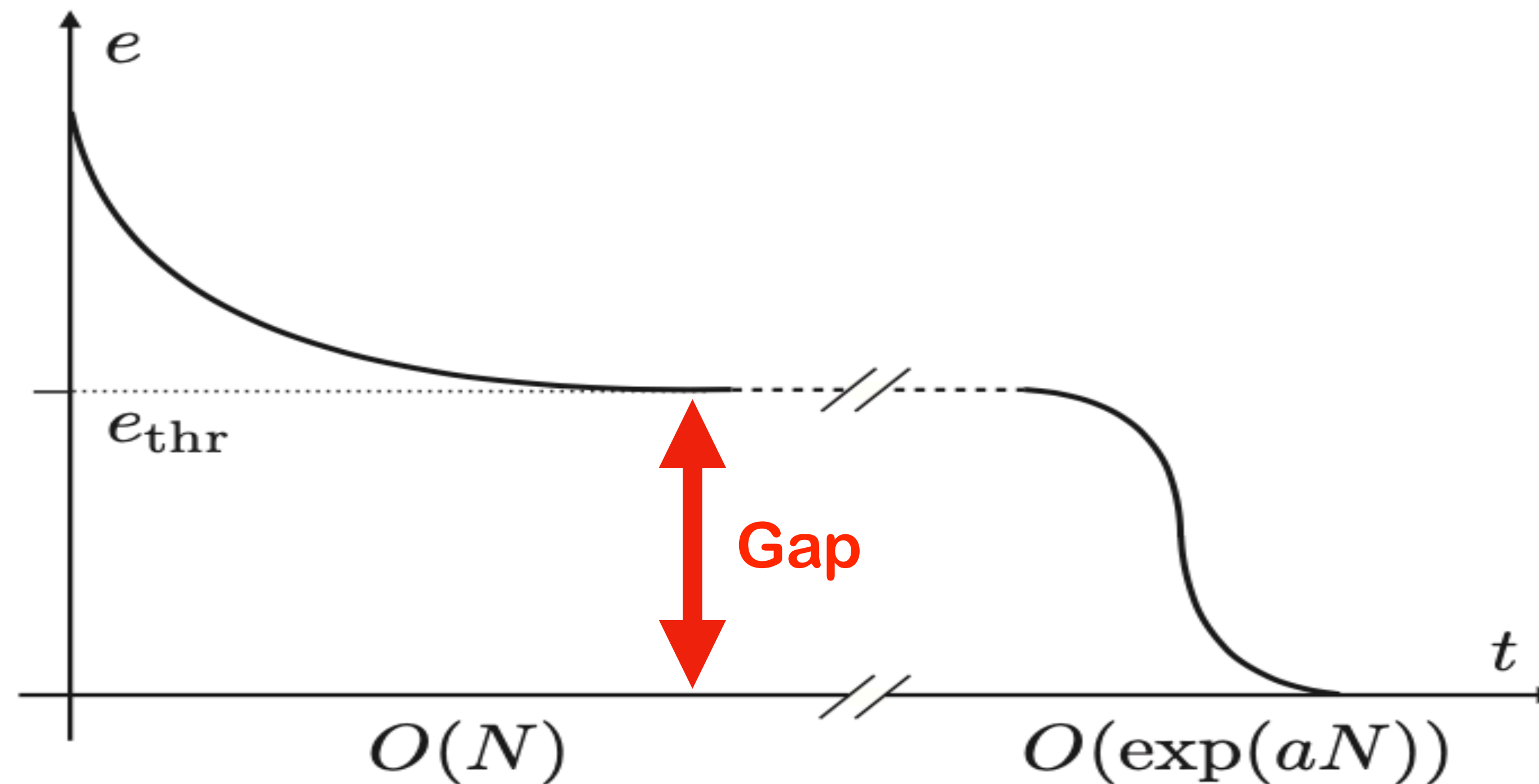
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 $\# \text{ states} \sim \exp(N \Sigma) \quad \Sigma : \text{complexity}$
- A key role is played by marginal states (i.e., flat minima) and entropic barriers



p-spin glasses have a dynamic transition

- For $T < T_d$, on polynomial timescales, the energy relaxes to the **threshold energy**, which is **above** the equilibrium energy



Numerical evidence for the dynamical transition

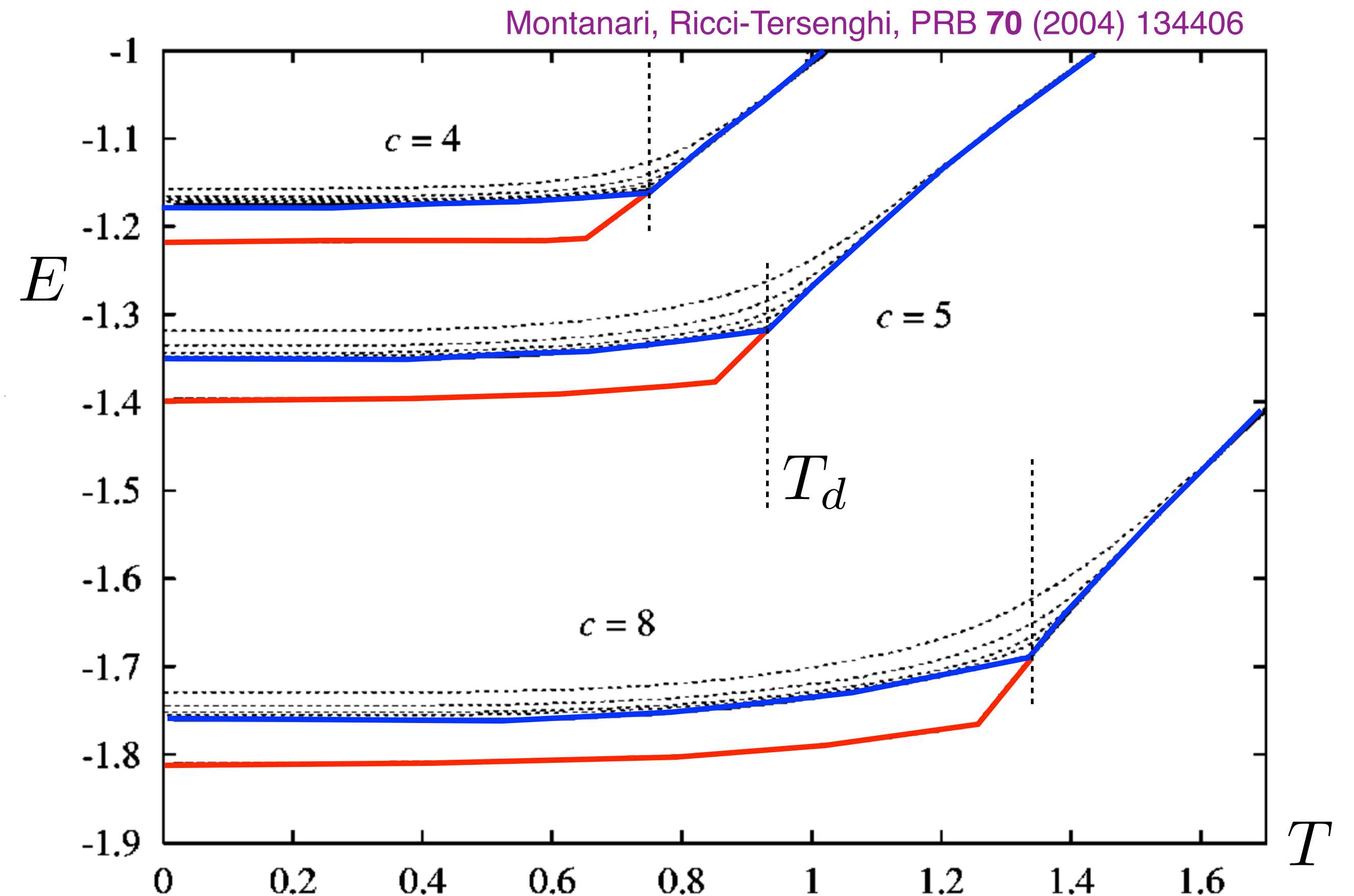
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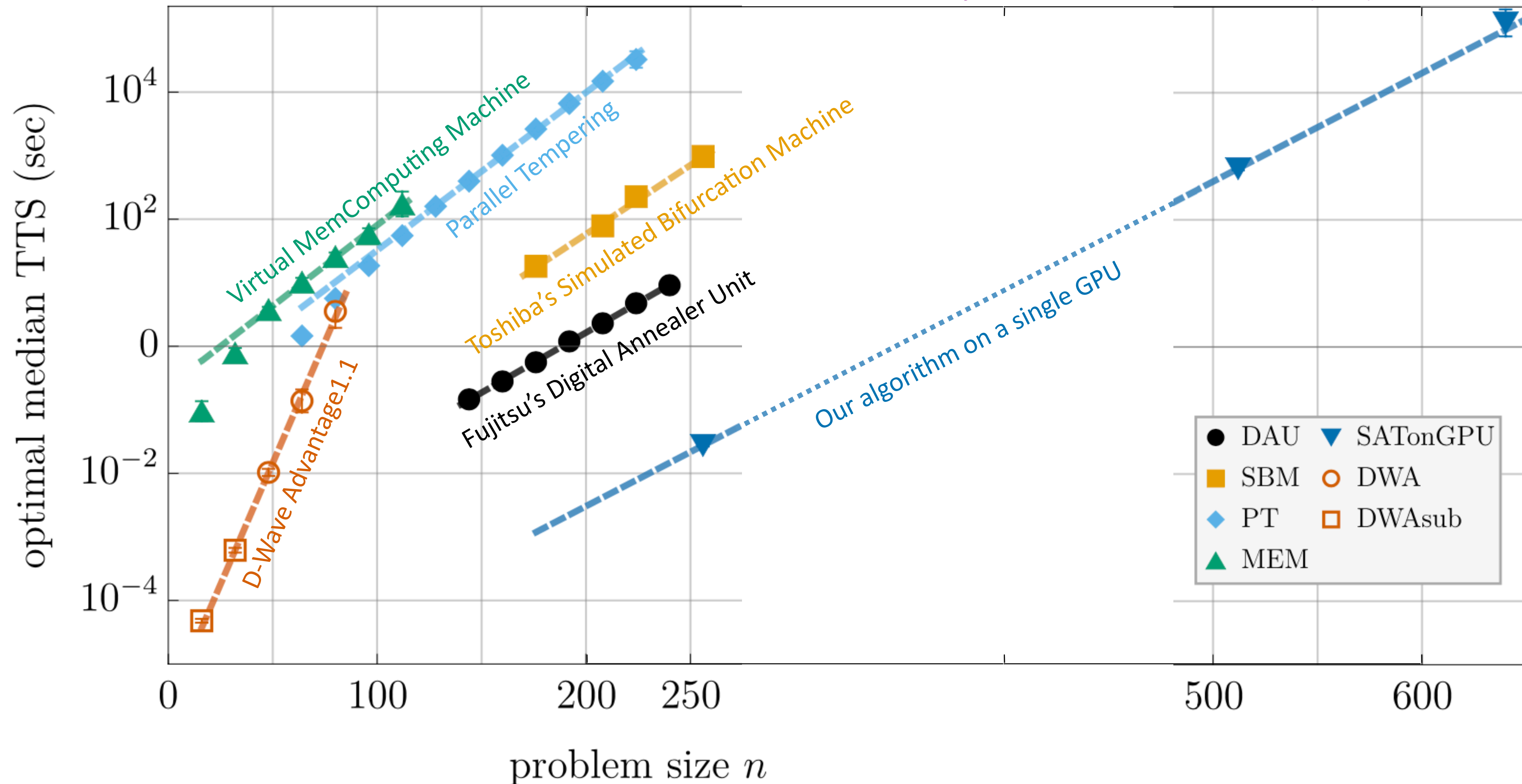
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- 3-spin glass on a c -regular sparse random hypergraph
- Ideal for numerical experiments:
 - efficient algorithms
SA (dashed lines)
 - analytical solutions
threshold energy
equilibrium energy



Optimizing a 3-spin model takes exponential time

Kowalsky et al, Quantum Sci. Technol. 7 (2022) 025008



Estimating the algorithmic complexity...

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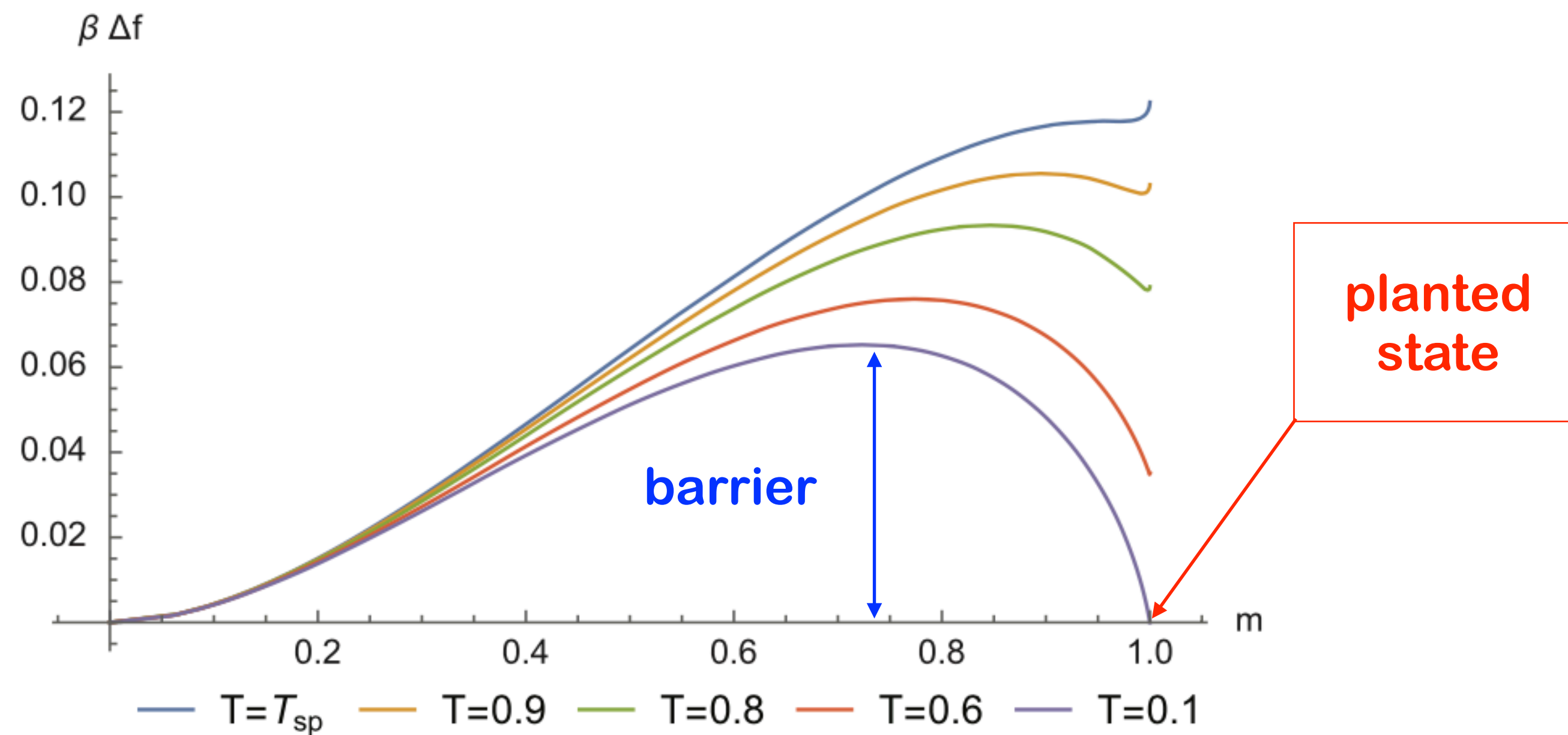
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Planted solution: $s_i = 1 \ \forall i$ (certified solution)

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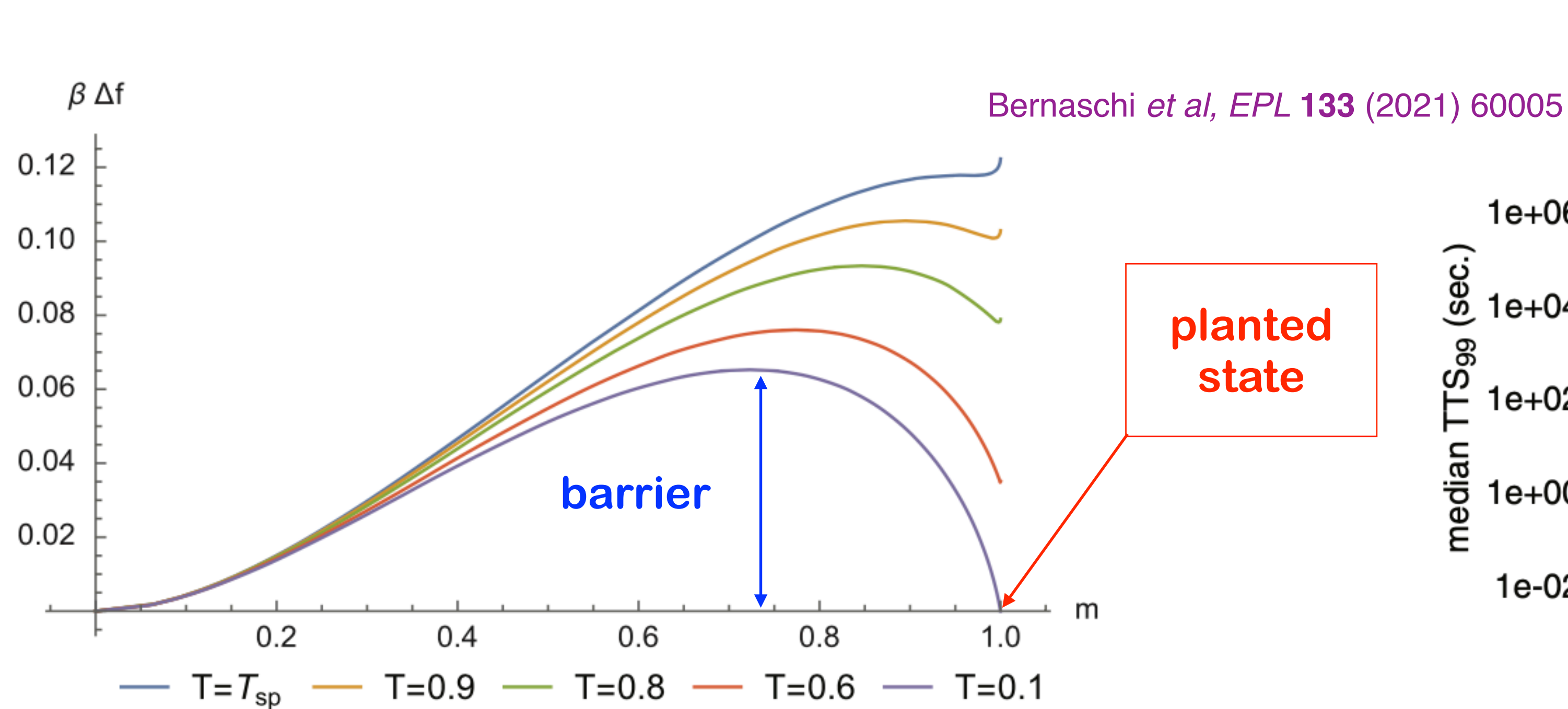


Estimating the algorithmic complexity...

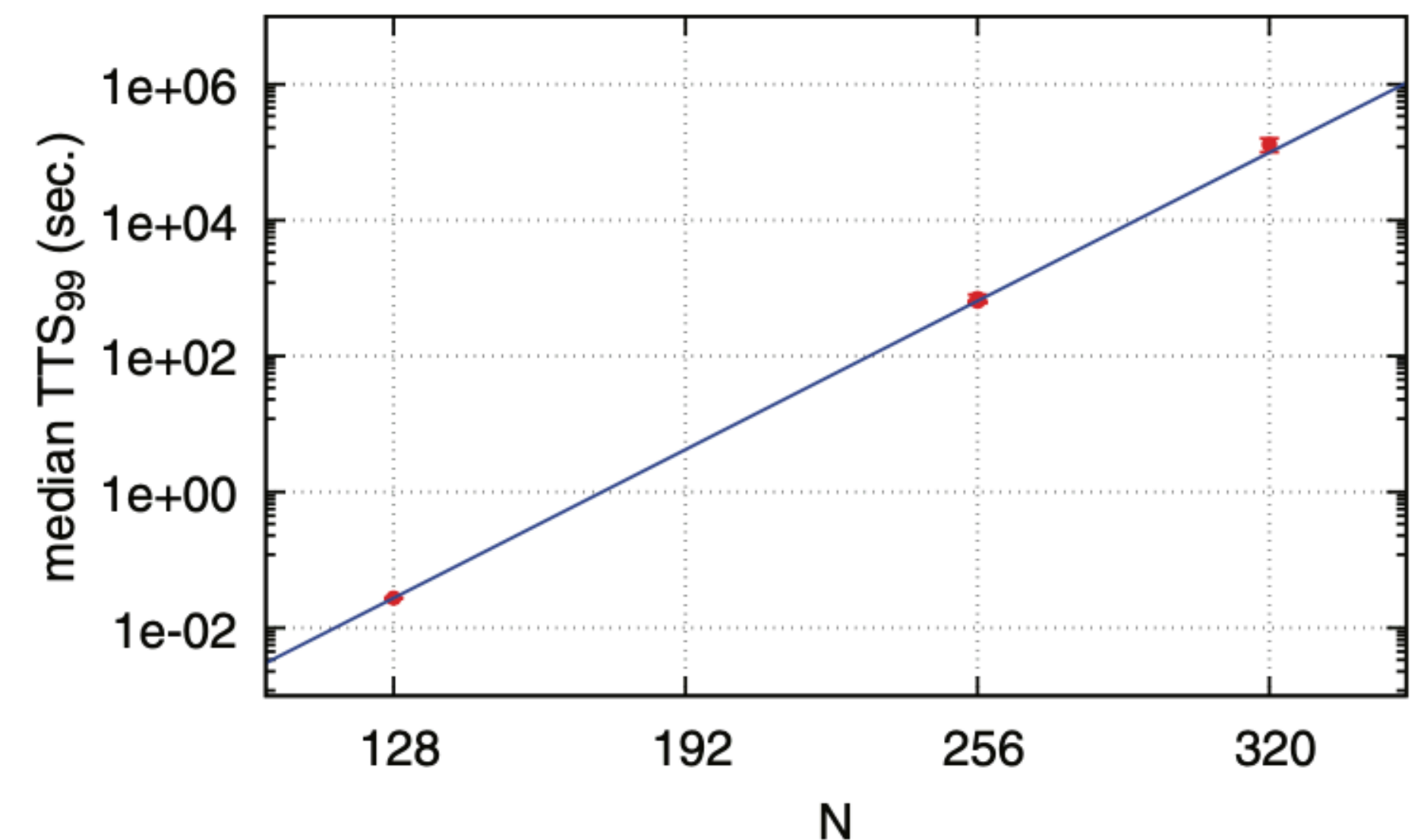
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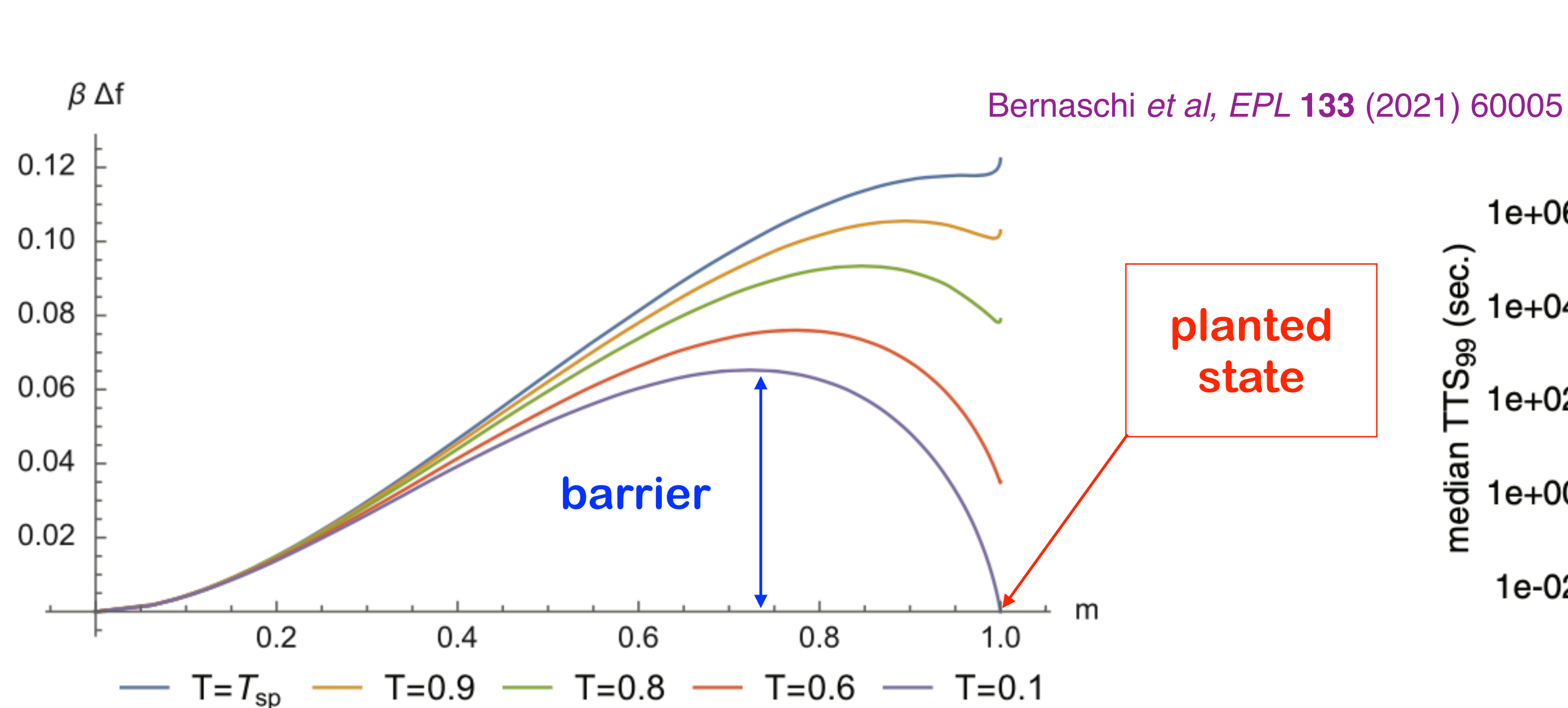


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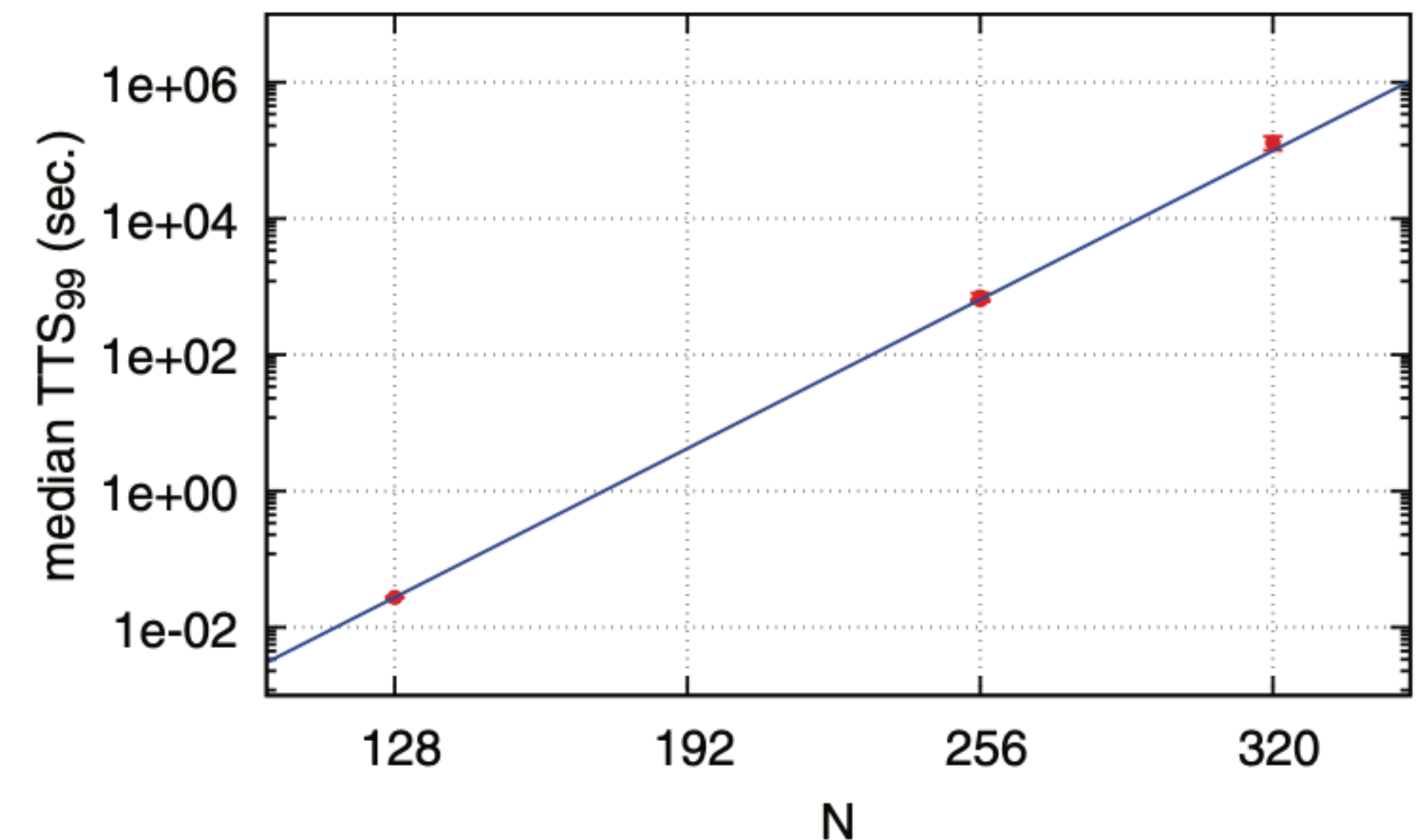
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Doing better than this exponential growth is a big challenge for quantum computing!

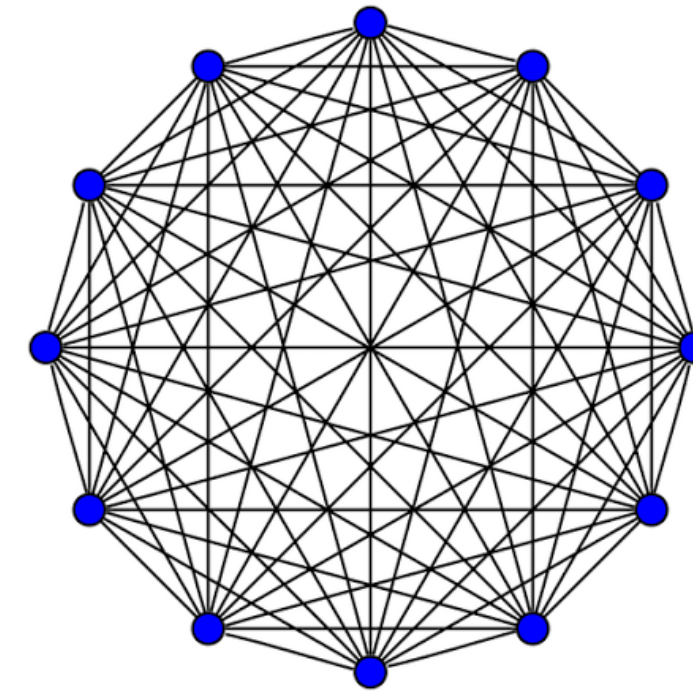
Spin glass models with a continuous phase transition

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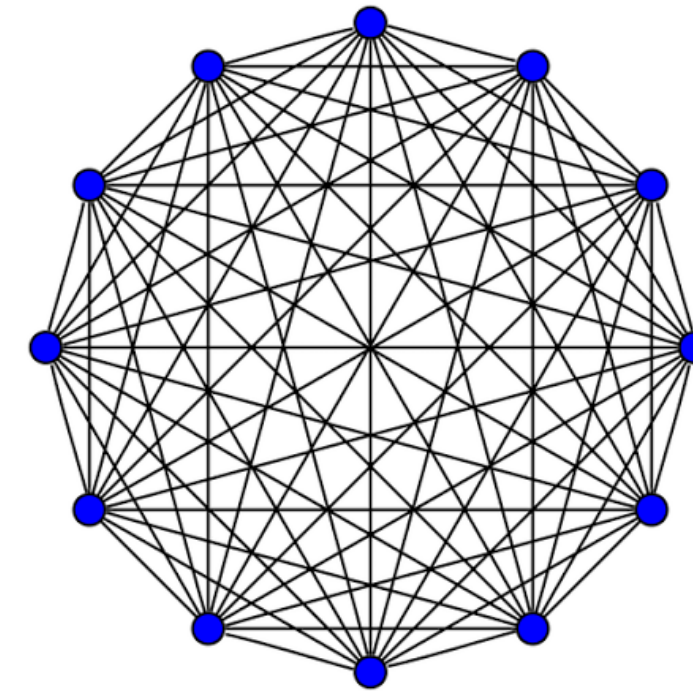
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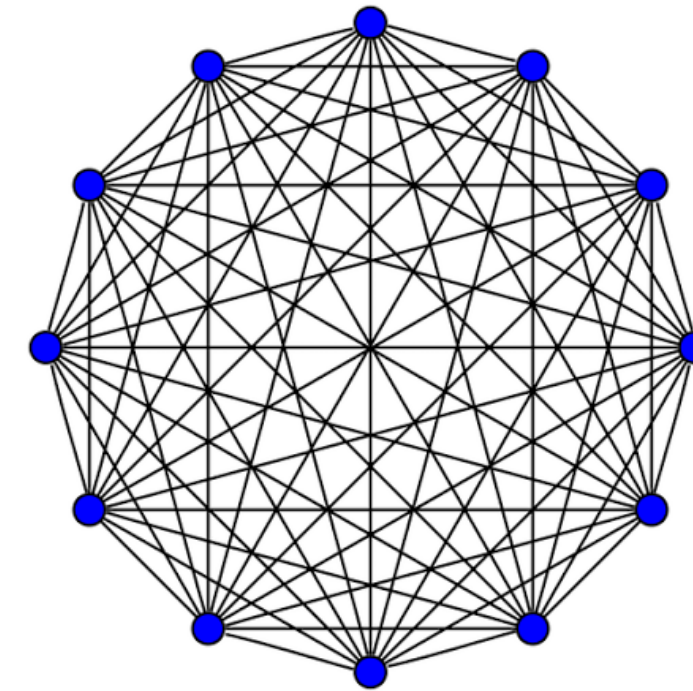


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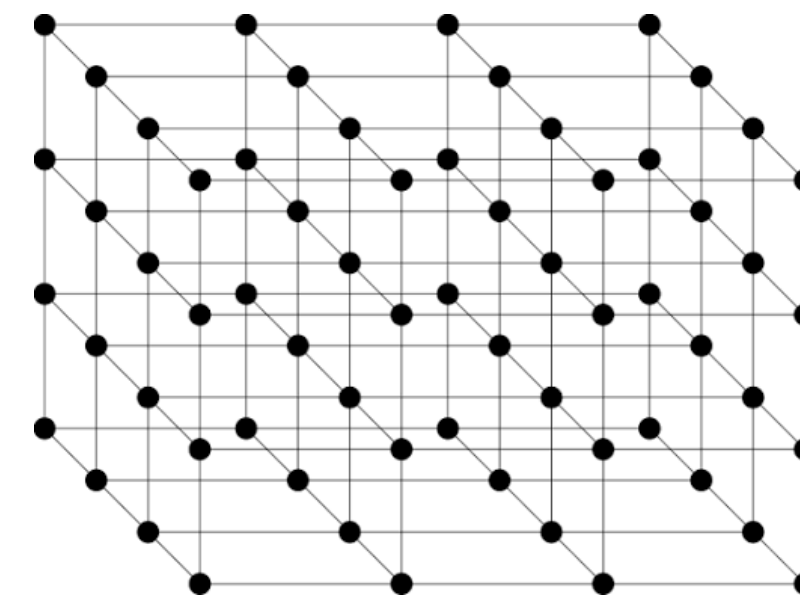


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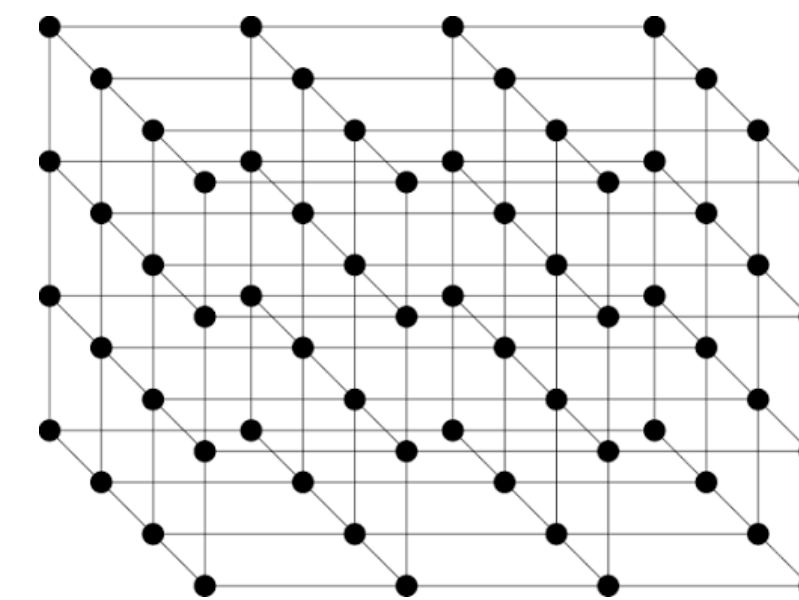
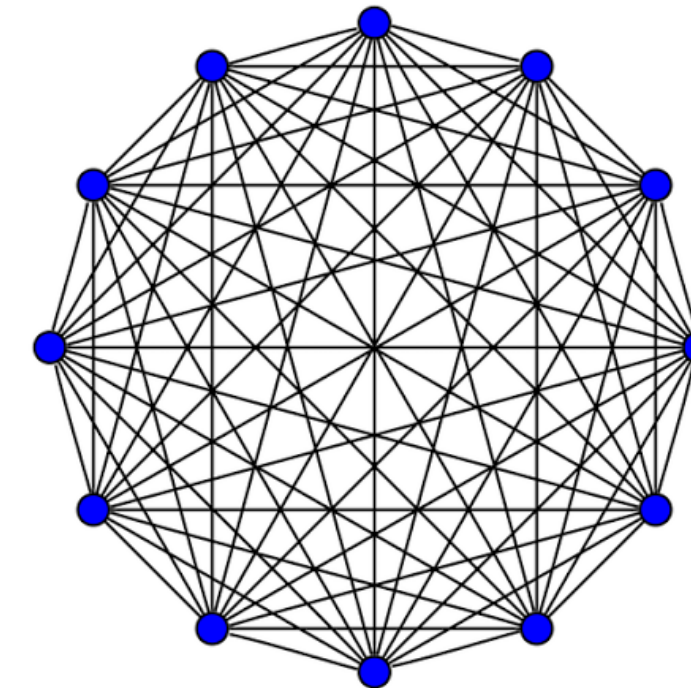
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Afternoon talk
by Francesco
Zamponi

Finding the ground state in the SK model

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- Let's try a very common classical algorithms (greedy and Simulated Annealing)

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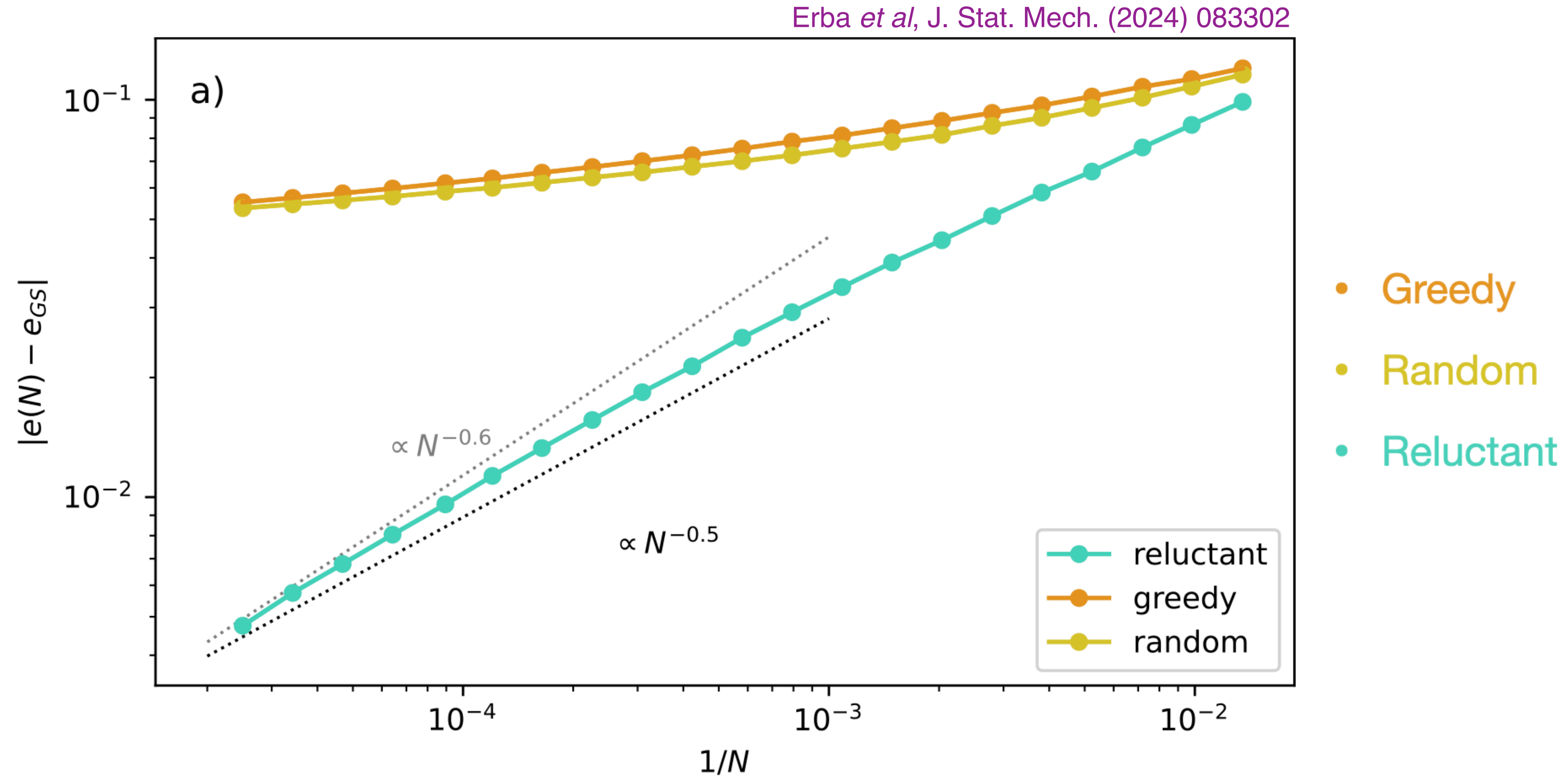
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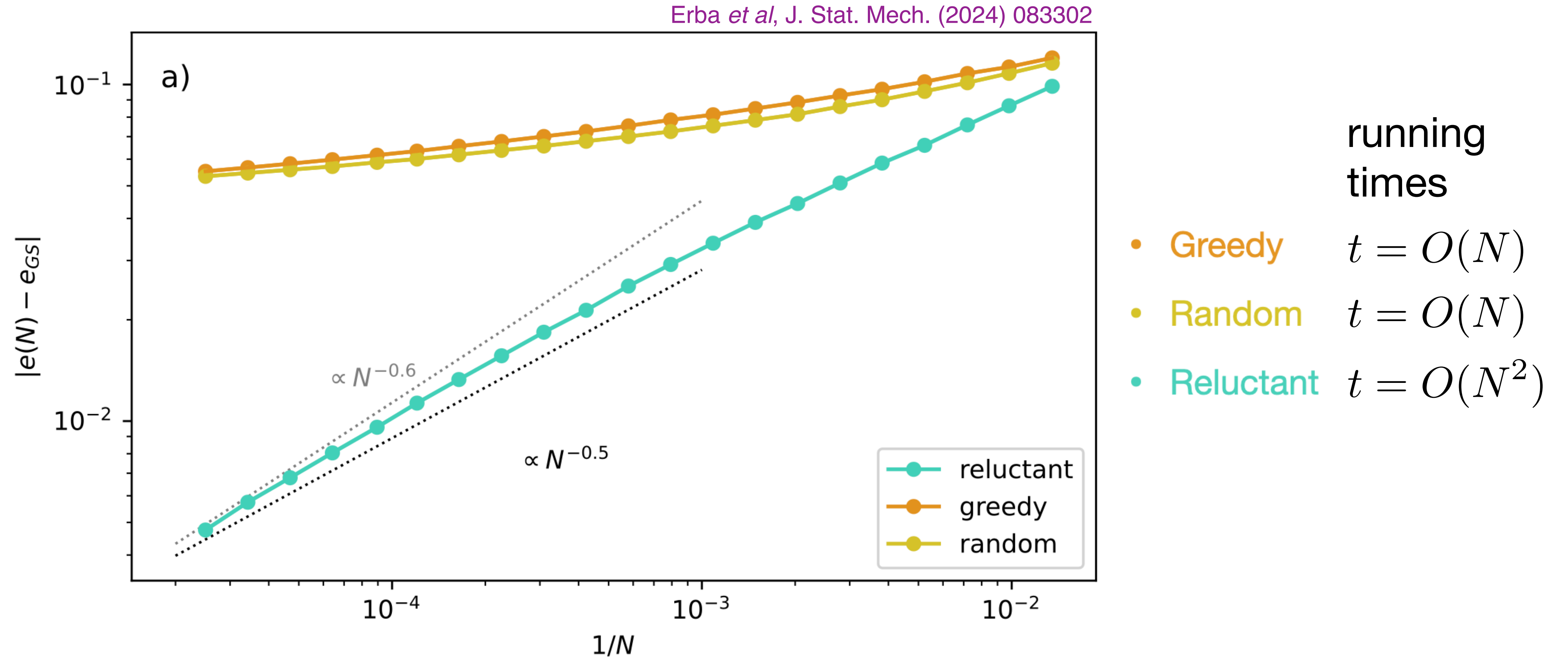
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- The algorithm stops when $\mathcal{U} = \emptyset$

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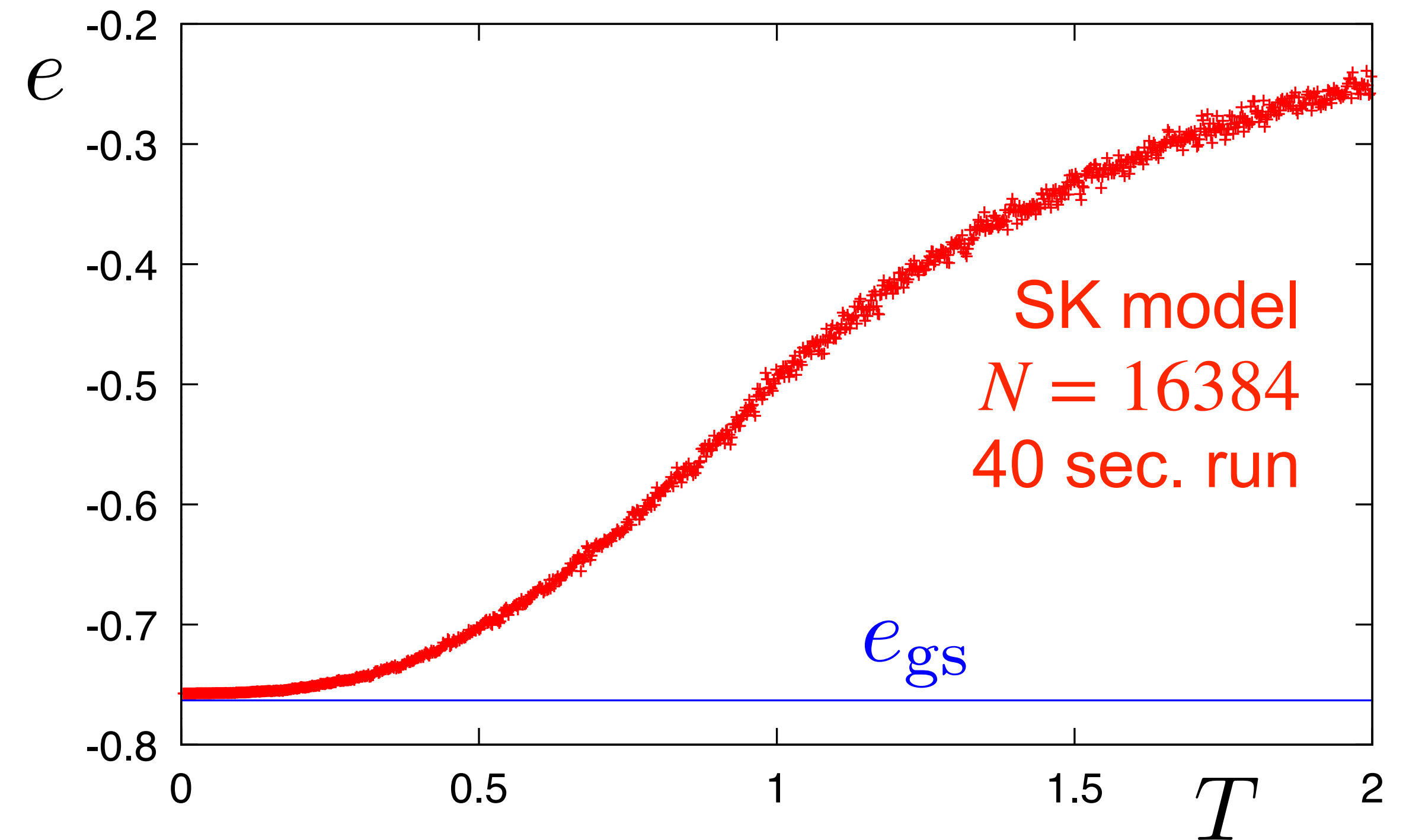
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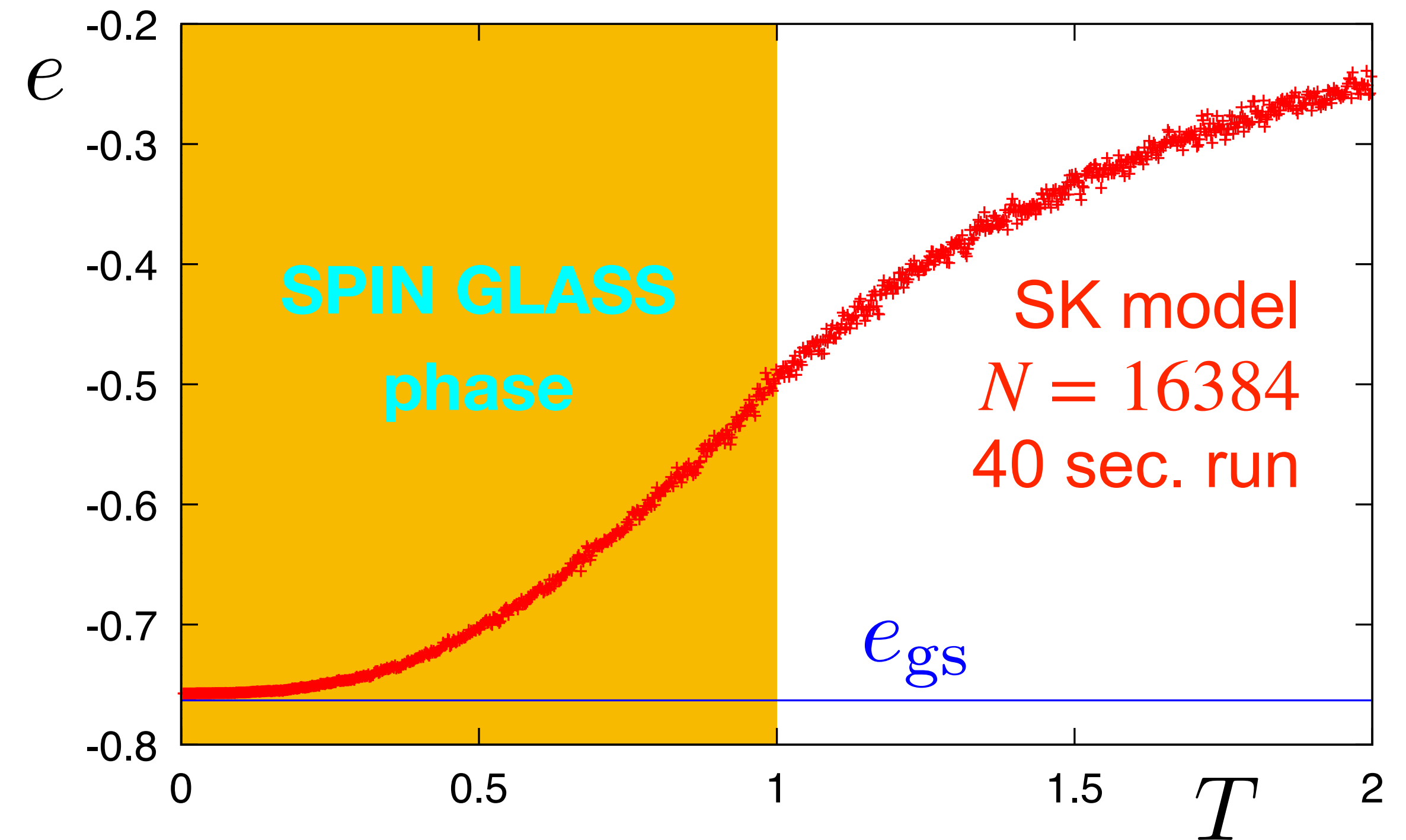
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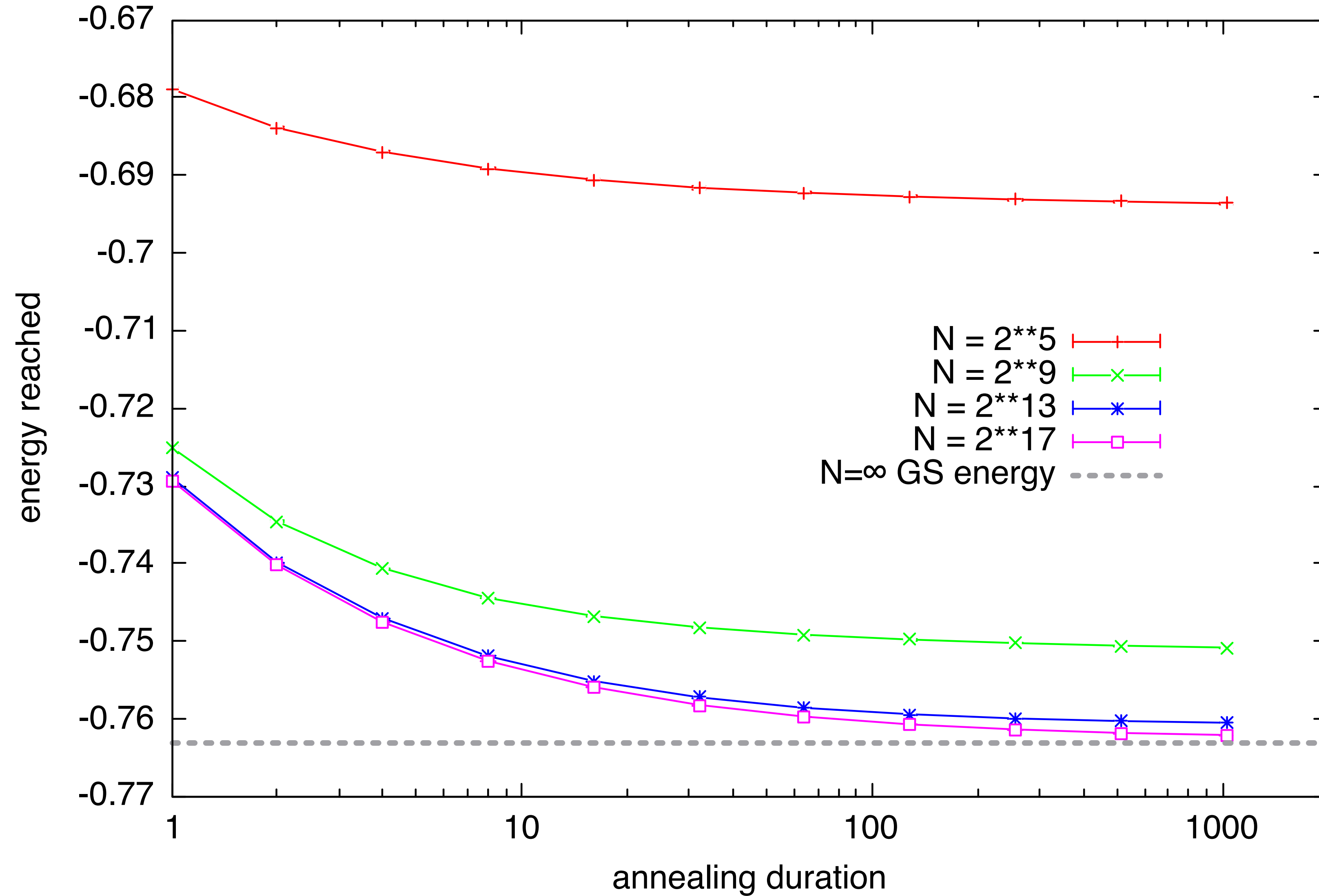


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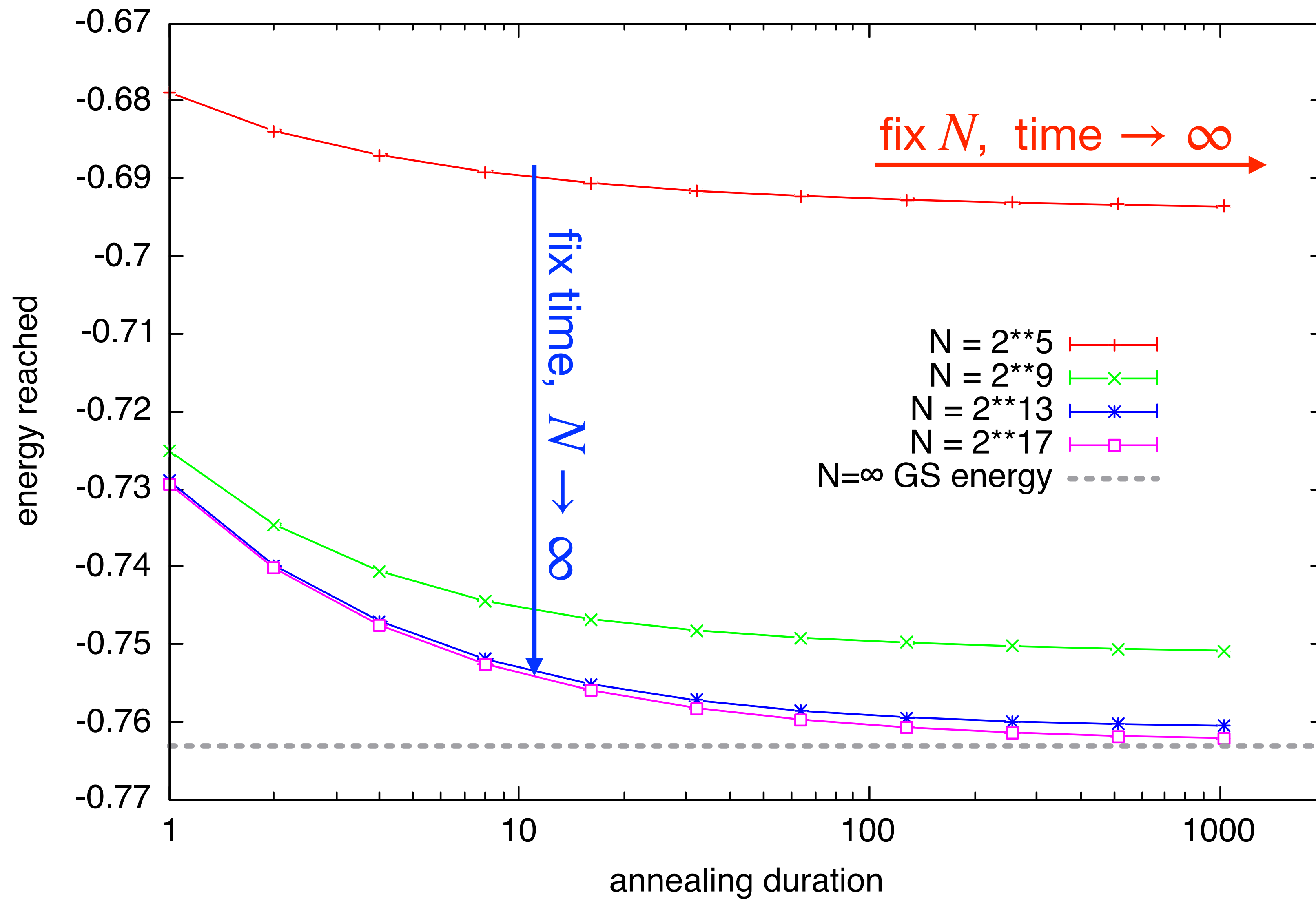
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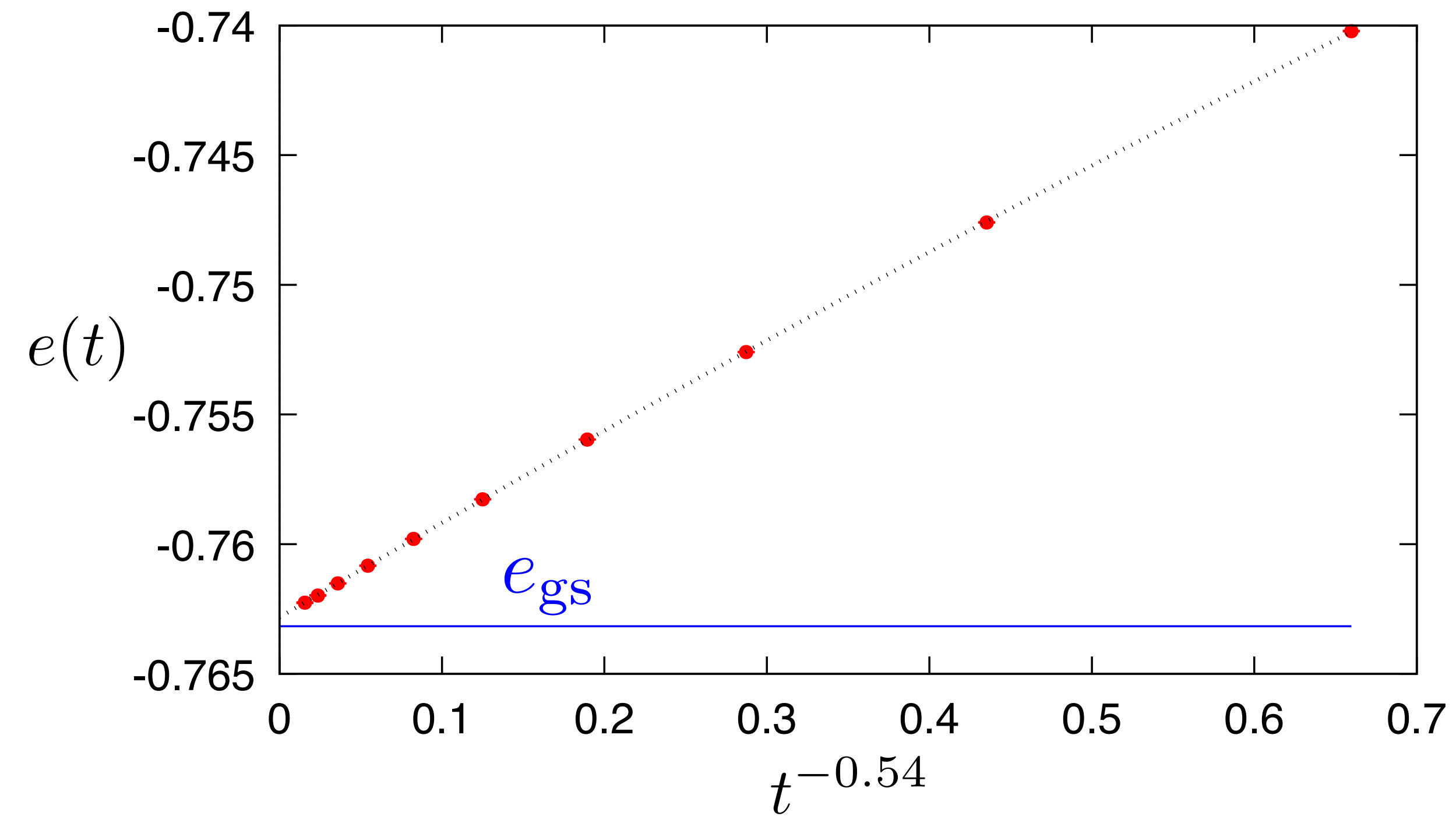
Performances of Simulated Annealing



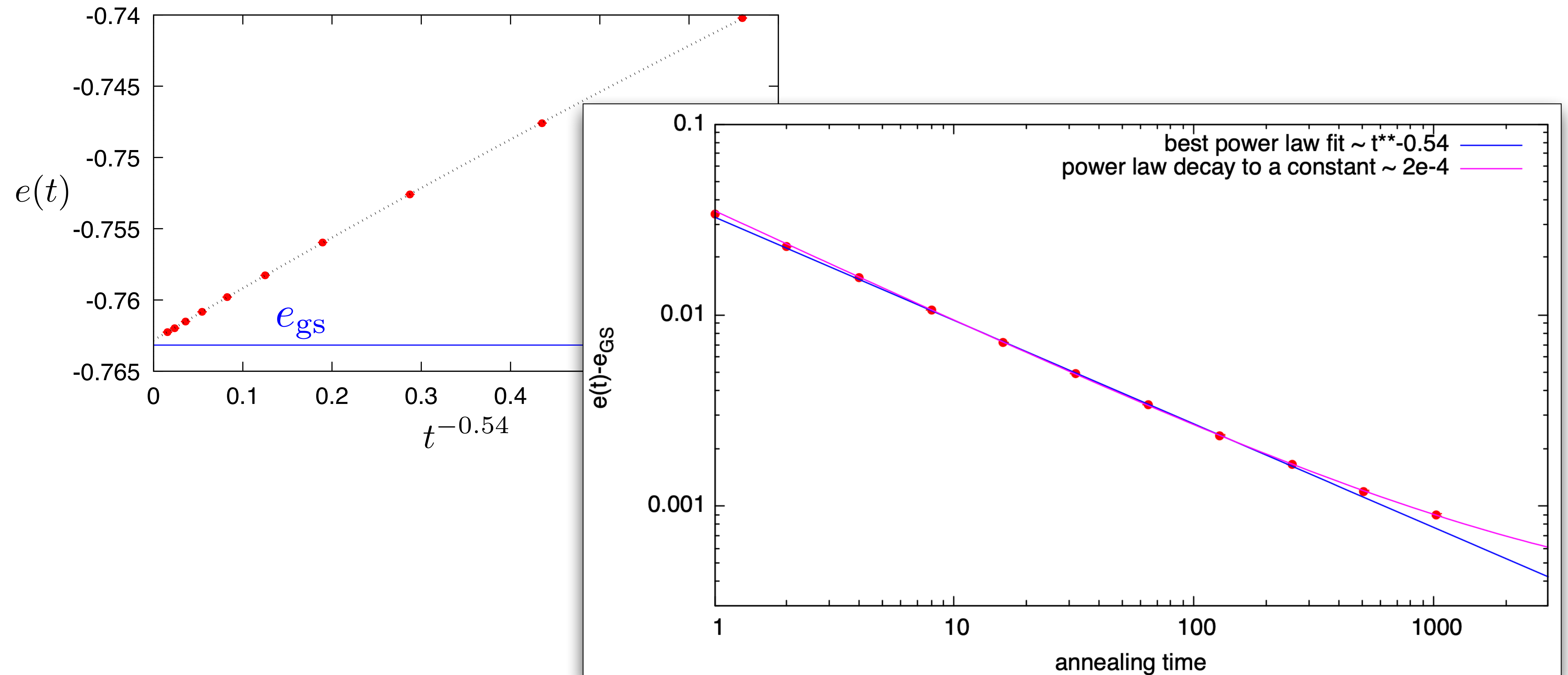
Two different limits



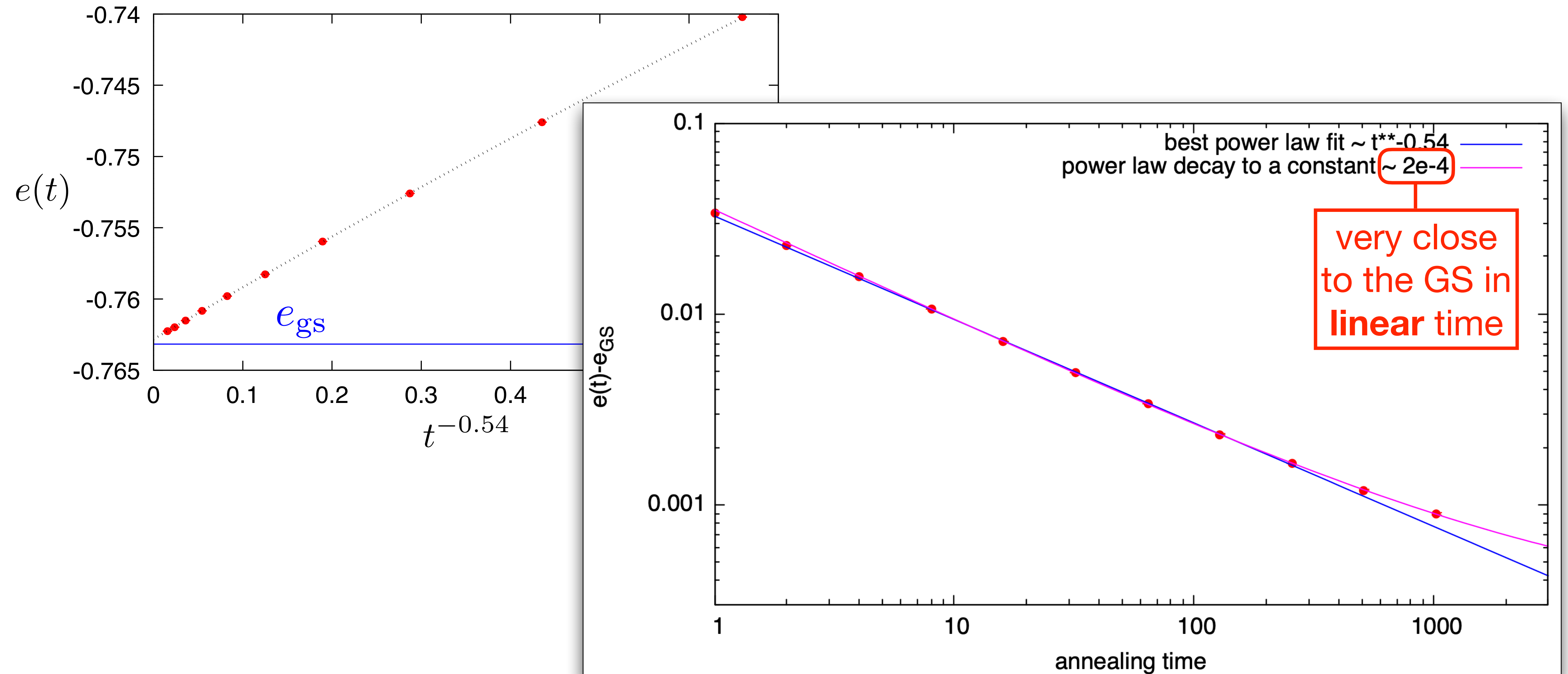
SA performances in $N = \infty$ limit (and linear running times)



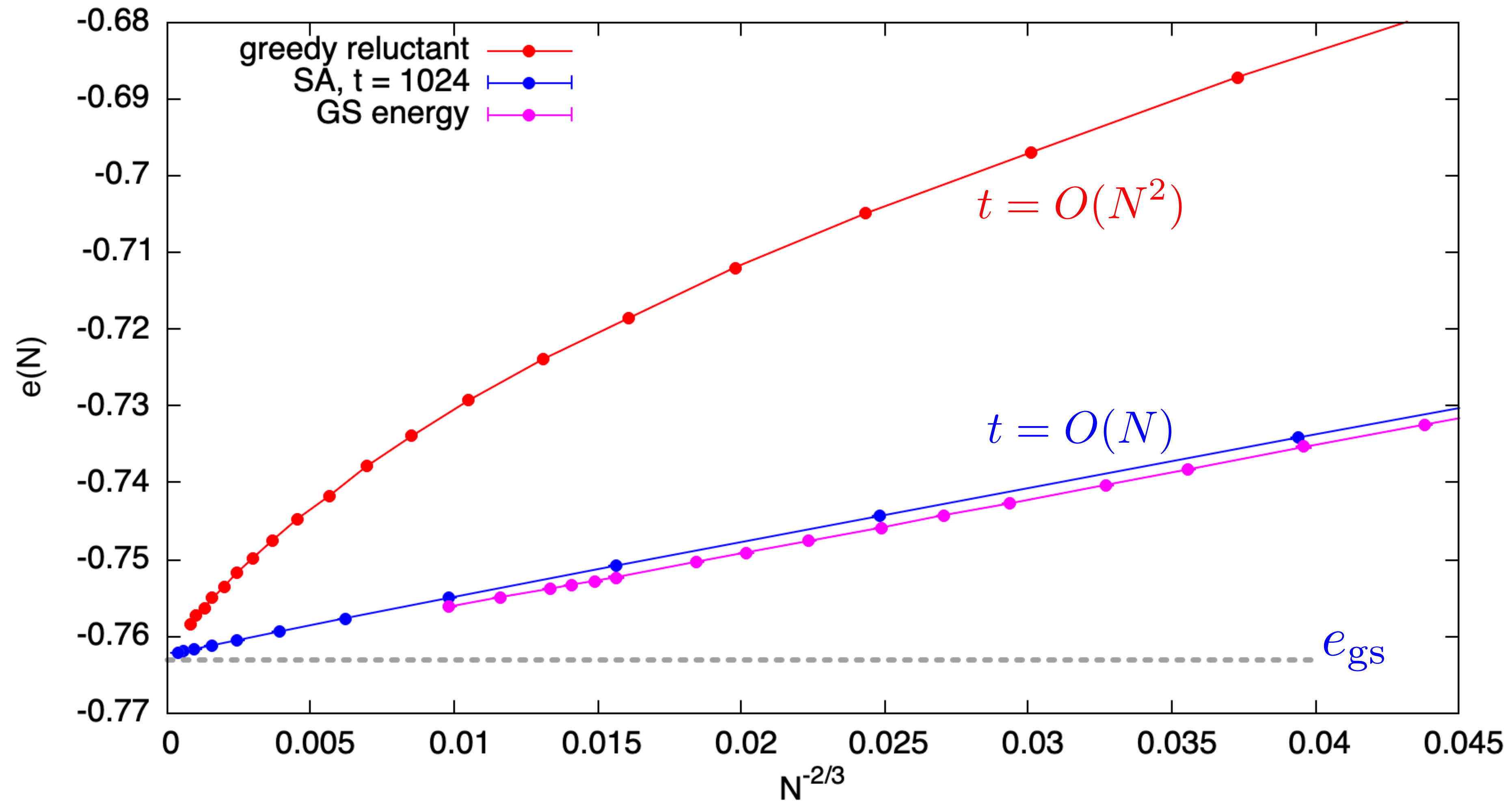
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Comparing algorithms performances for finite N



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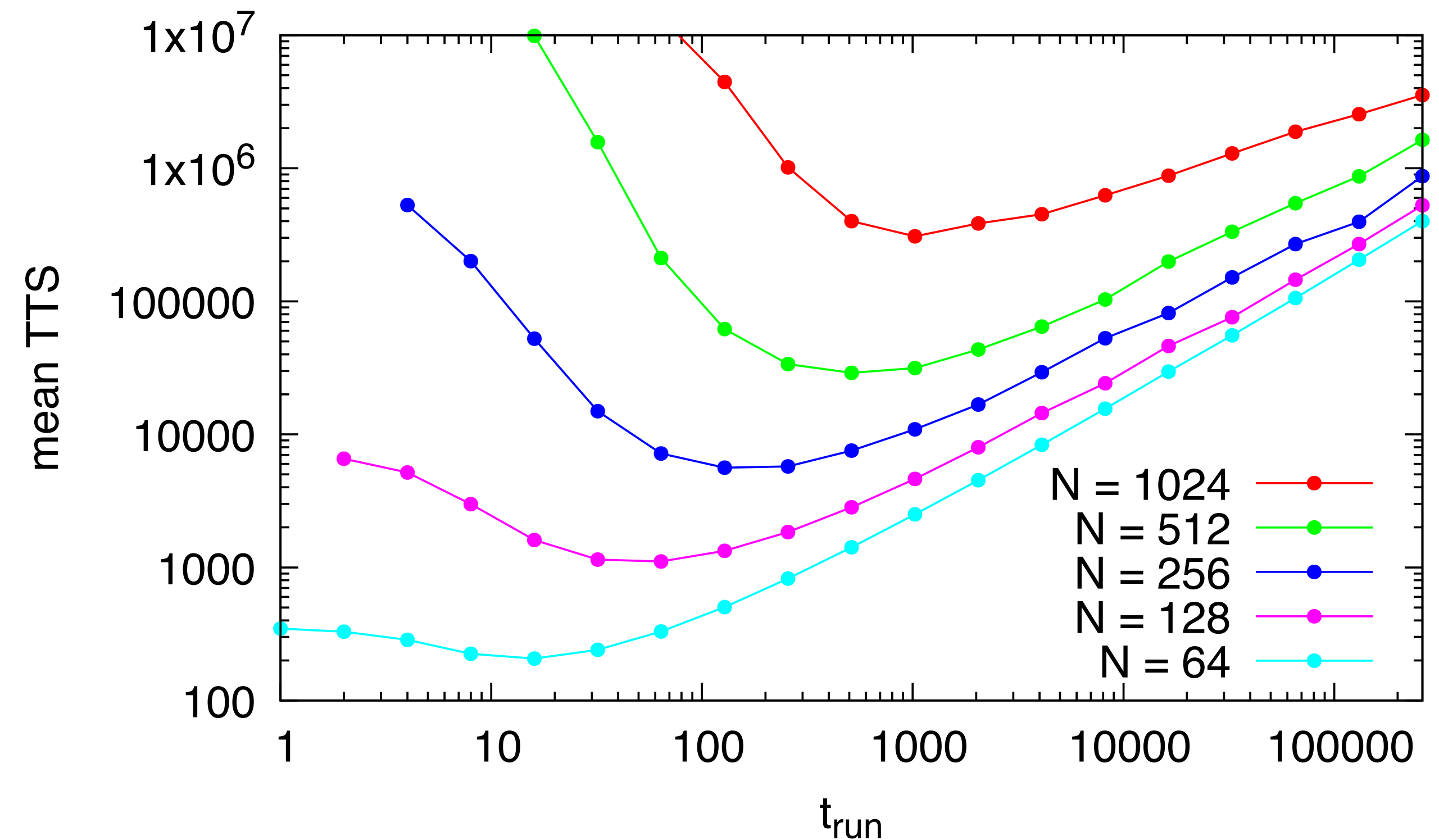
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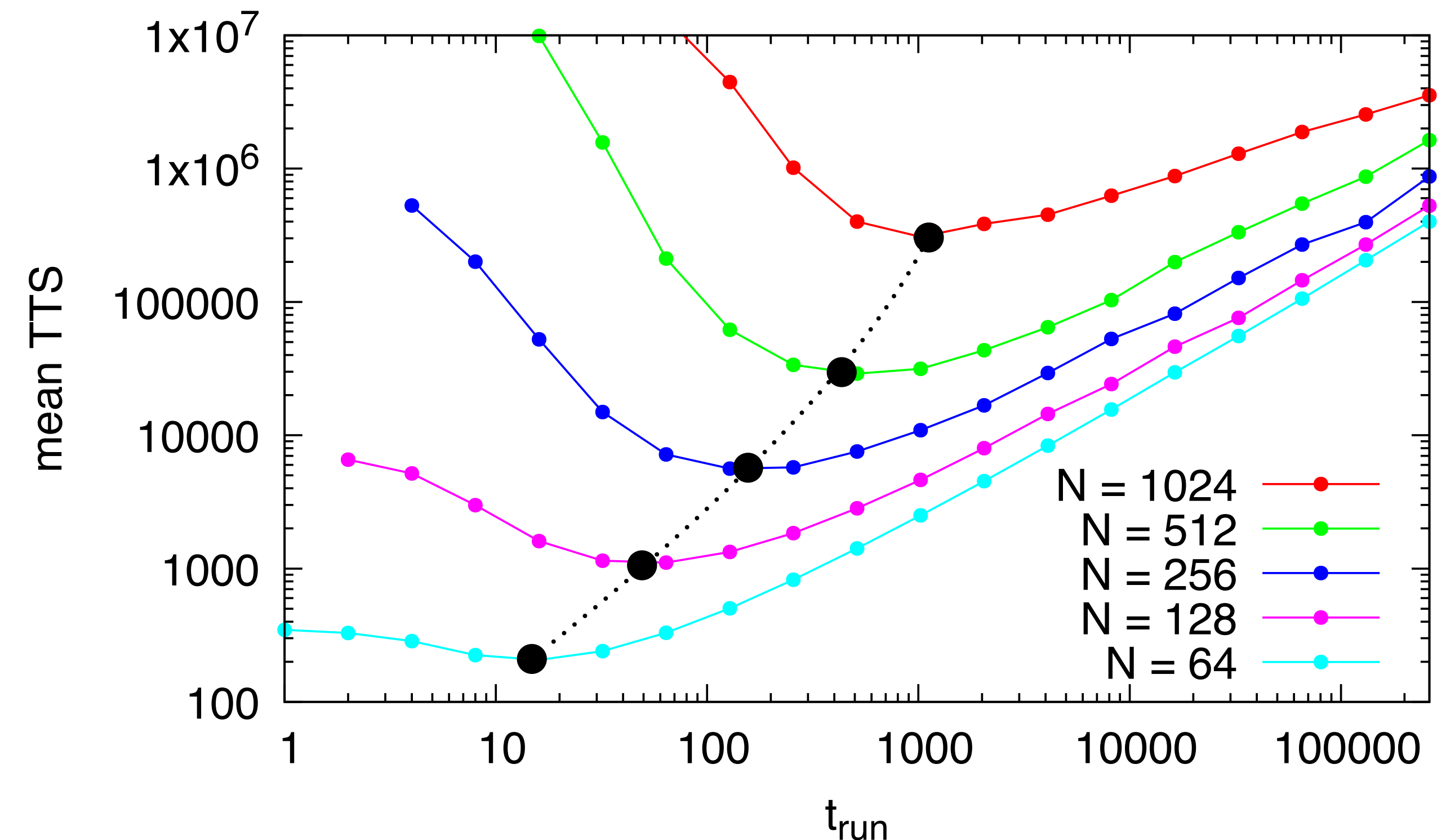
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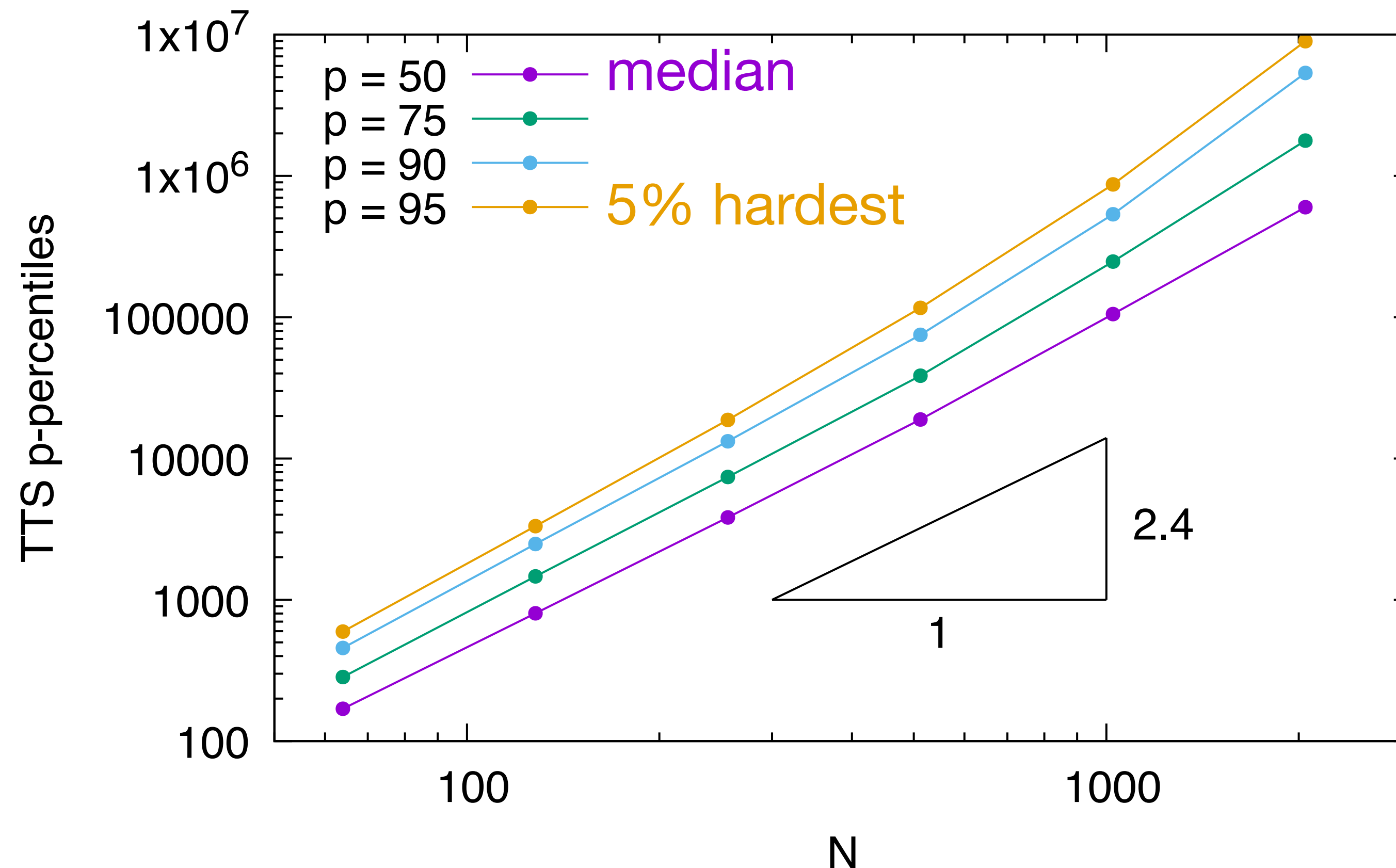
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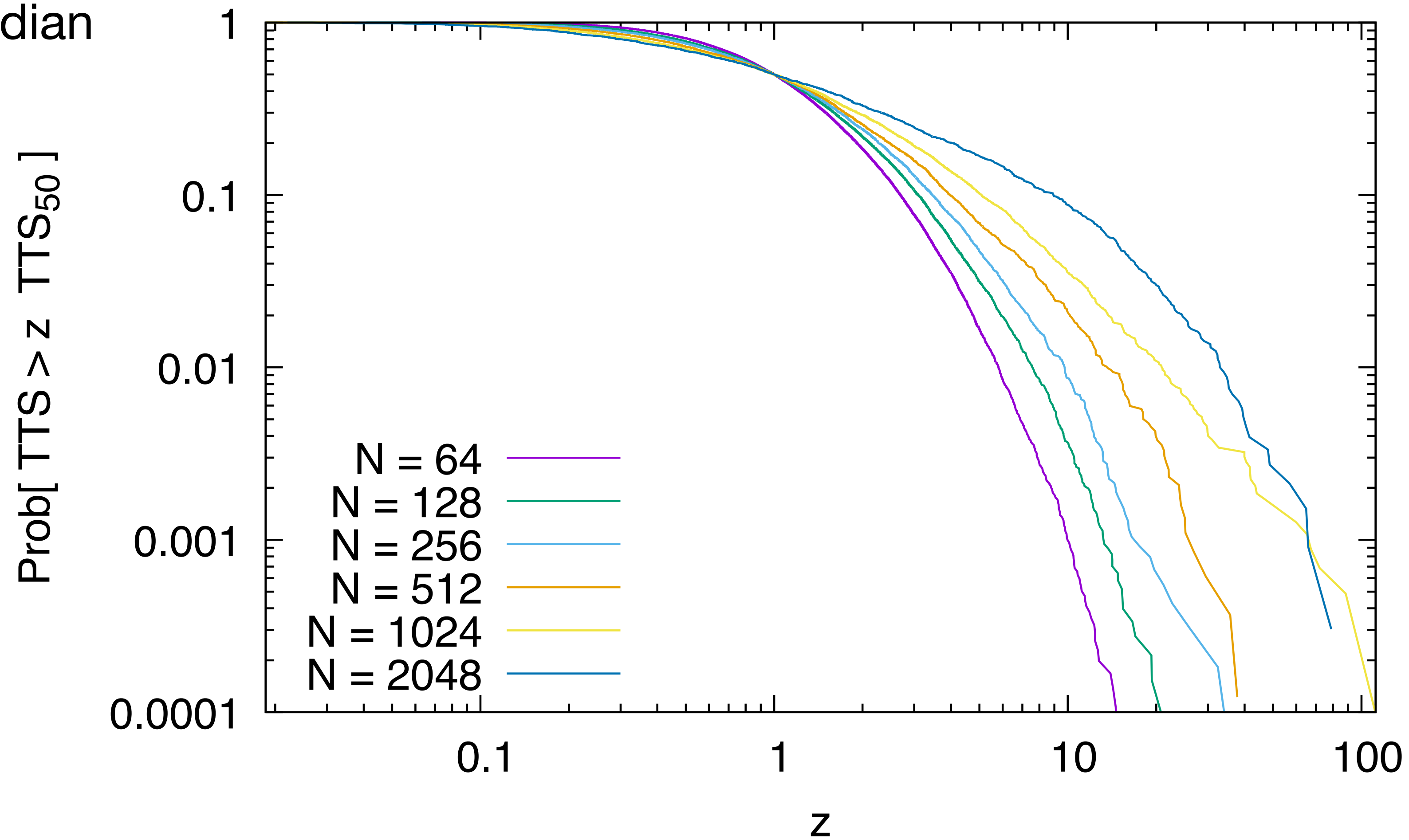
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- Fixed annealing time $t(N)$, run SA to solve thousands of samples
- Estimate the computational complexity of **typical** and **hardest** samples



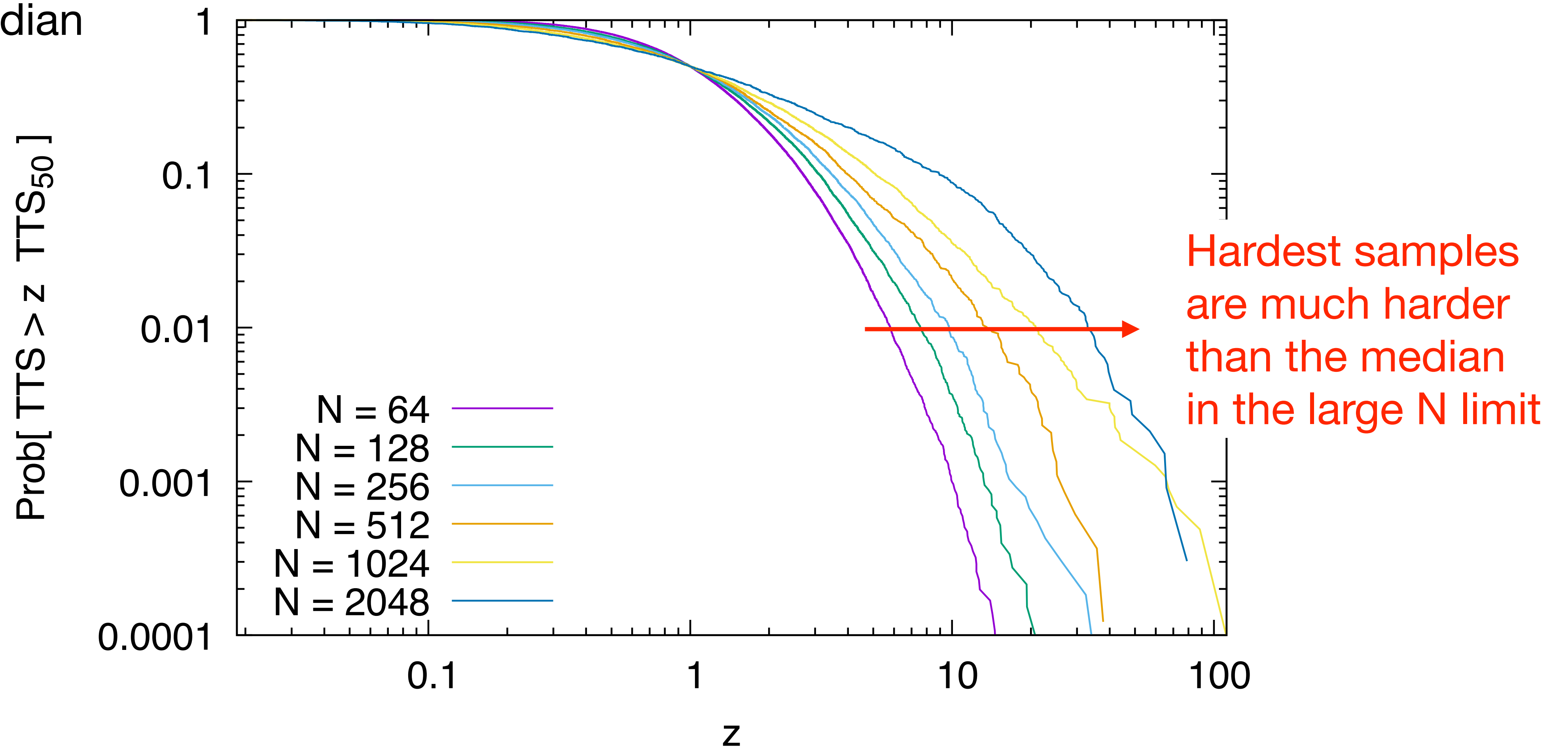
Large deviations analysis to estimate the actual complexity

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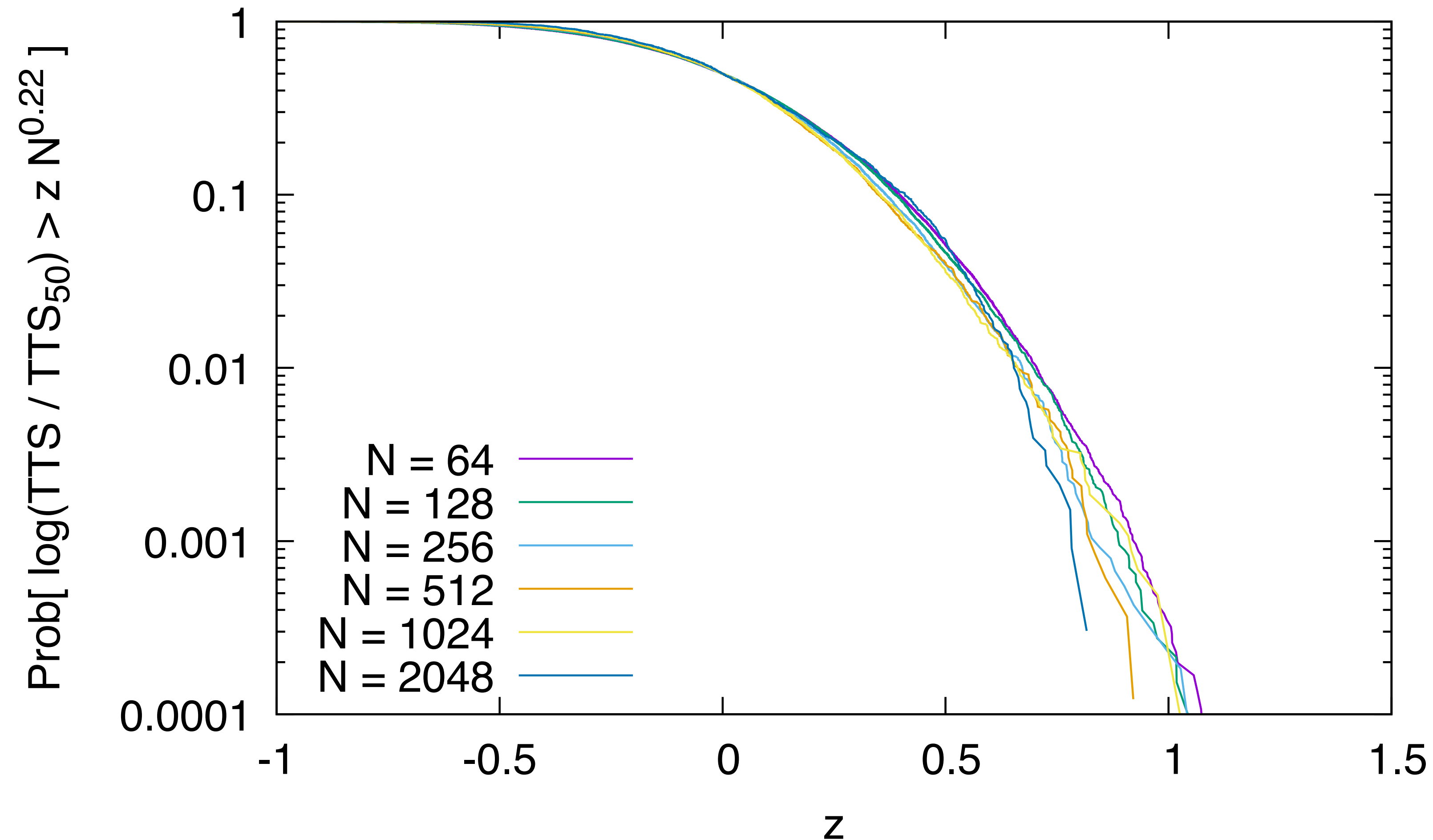
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In the SK model, at low T, free-energy barriers grow like $N^{0.22} \implies t \sim \exp(N^{0.22})$



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 - Large sizes and many samples are needed (otherwise look polynomial...)

Thank you!